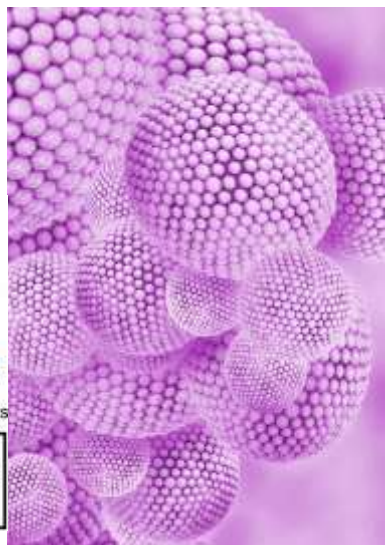
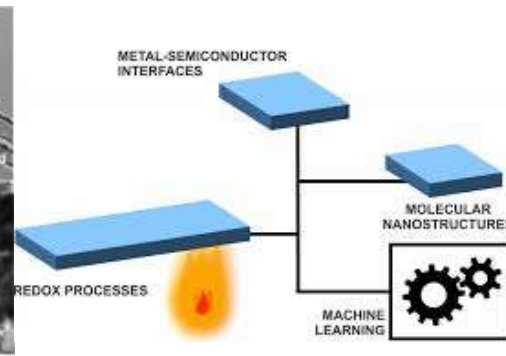
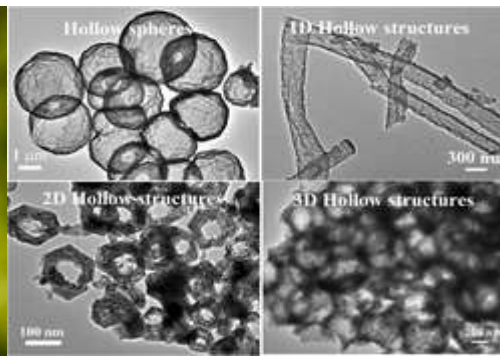
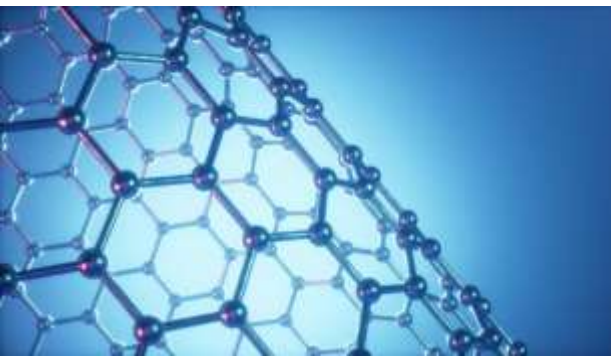
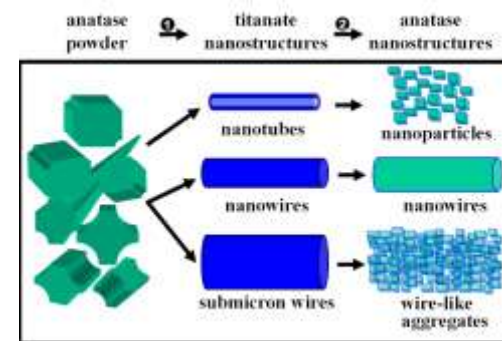
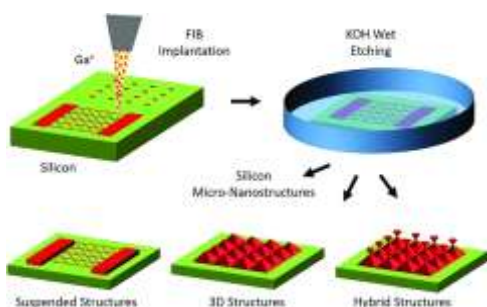




Indian Institute of Information Technology, Surat

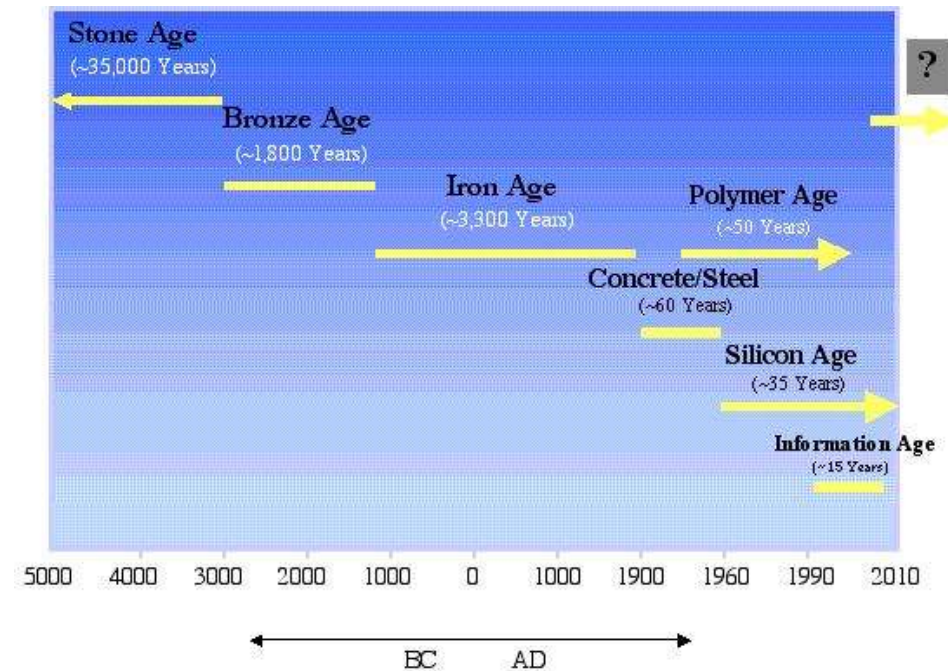
Study of the Properties of the Nanomaterials

Dr. Nishad G. Deshpande



History of Physics, Materials Science & Engineering

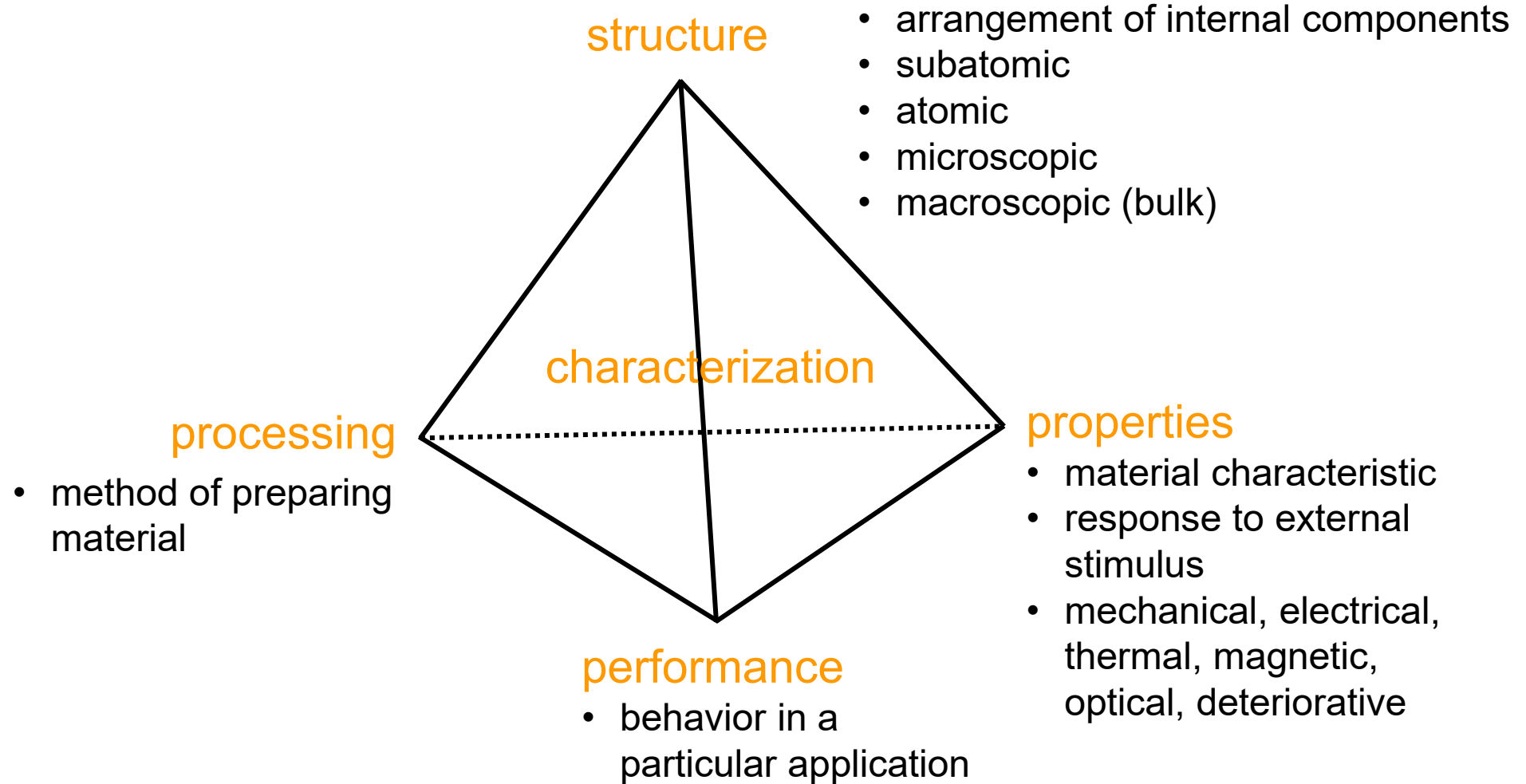
- materials closely connected our culture
- the development and advancement of societies are dependent on the available materials and their use
- **Early civilizations** is connected with **Materials development**



- initially **natural materials**
- develop techniques to produce materials with superior qualities (heat treatments and addition of other substances)

MATERIALS SELECTION!

Materials Science and Engineering



Classification of Materials

Metals

- good conductors of electricity and heat
- lustrous appearance
- susceptible to corrosion
- strong, but deformable



Ceramics & Glasses

- thermally and electrically insulating
- resistant to high temperatures and harsh environments
- hard, but brittle



Polymers

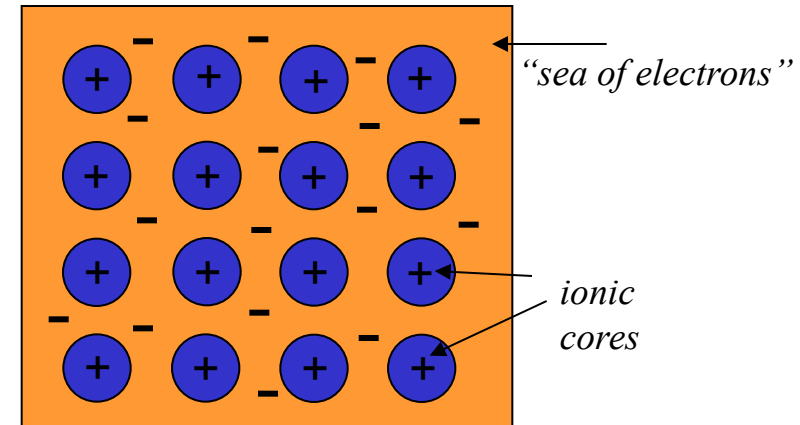
- very large molecules
- low density, low weight
- maybe extremely flexible



Metals

Metallic Bond

- one, two, or three valence electrons
- valence electrons free to drift through the entire material forming a “*sea of electrons*” surrounding net positive *ionic cores*
- non-directional bond



GROUP		Periodic Table of the Elements																		VIII		
IA																				VIII		
1	H																			2	He	
2	Li	Be																			10	Ne
3	Na	Mg																			18	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr				
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe				
6	Cs	Ba		Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn				
7	Fr	Ra																				

Properties

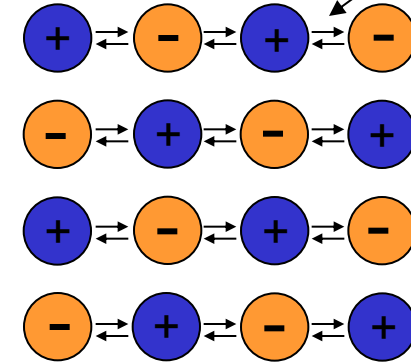
- good conductors of electricity and heat
- lustrous appearance
- susceptible to corrosion
- strong, but deformable

Ceramics and Glasses

Ionic Bond

- composed of metallic and non-metallic elements
- metallic elements give up valence electrons to non-metallic elements
- all atoms have filled “inert gas” configuration
- ionic solid
- non-directional bond

Coulombic bonding force



GROUP IA		Periodic Table of the Elements																VIII		
1	H																	2	He	
2	Li	Be																	10	Ne
3	Na	Mg																	18	Ar
4	K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr		
5	Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe		
6	Cs	Ba																	86	Rn
7	Fr	Ra																		
			57	La	Ce	Pr	Nd	Pm	62	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	
			89	Ac	Th	Pa	U	Np	94	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	

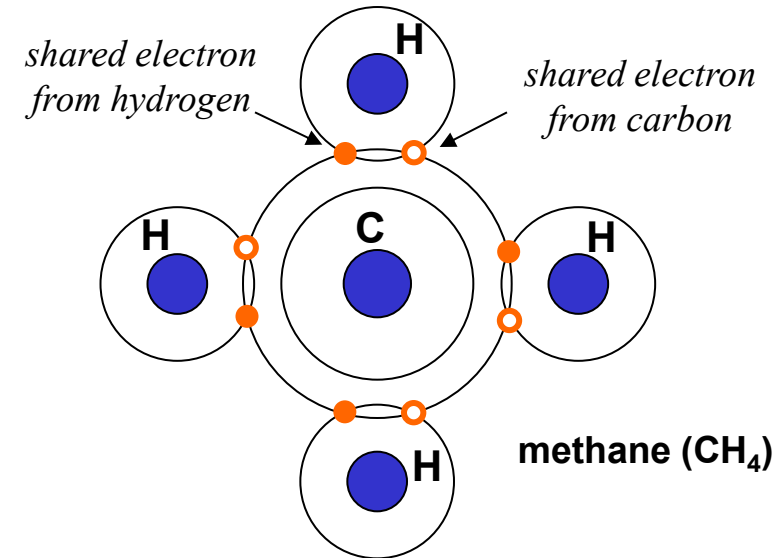
Ceramics & Glasses

- thermally and electrically insulating
- resistant to high temperatures and harsh environments
- hard, but brittle

Polymers

Covalent Bond

- electrons are shared between adjacent atoms, each contributing at least one electron
- shared electrons belong to both atoms
- directional bond

[illegible]

Polymers

- very large molecules
- low density, light weight materials
- maybe extremely flexible

Why study materials?

- applied scientists or engineers must make material choices
- materials selection
 - in-service performance
 - deterioration
 - economics

BUT...really, everyone makes material choices!

aluminum



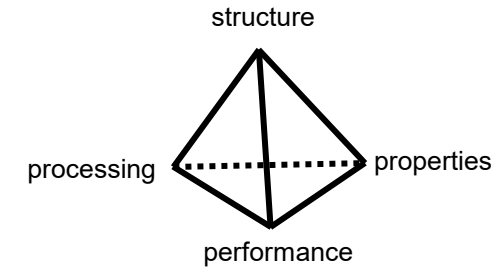
glass



plastic



Levels of Structure



← **STRUCTURE** (length scale)

↓
Sub-atomic

< 0.2 nm
1 nm = ?

↓
Atomic

0.2-10 nm

↓
Microscopic

1-1000 μm

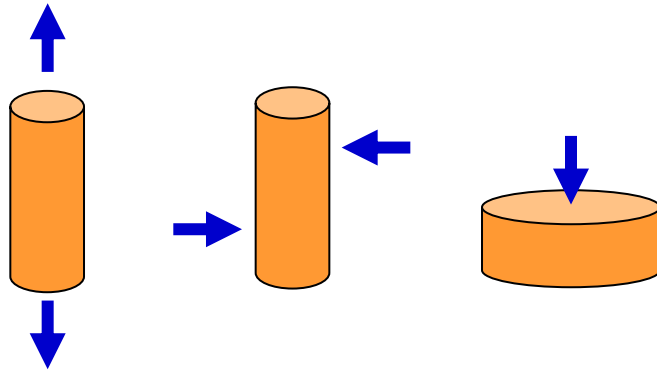
↓
Bulk

> 1 mm

Bulk Properties

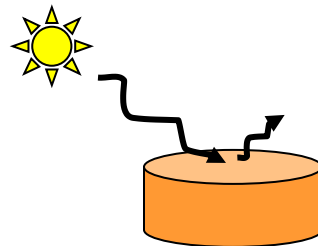
Mechanical:

elastic modulus
shear modulus
hardness



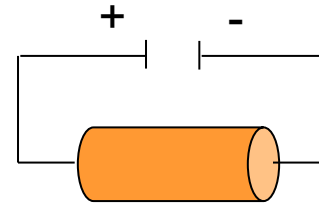
Optical:

reflectivity
absorbance
emission



Electrical:

conductivity
resistivity
capacitance

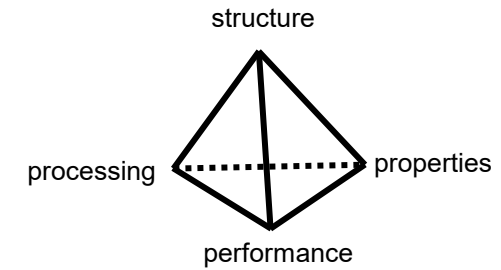


Thermal:

thermal expansion
heat capacity
thermal conductivity



Levels of Structure

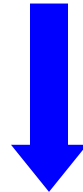


← **STRUCTURE** (length scale)



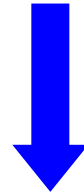
Sub-atomic

< 0.2 nm
1 nm = ?



Atomic

0.2-10 nm



Microscopic

1-1000 μm

Microstructure

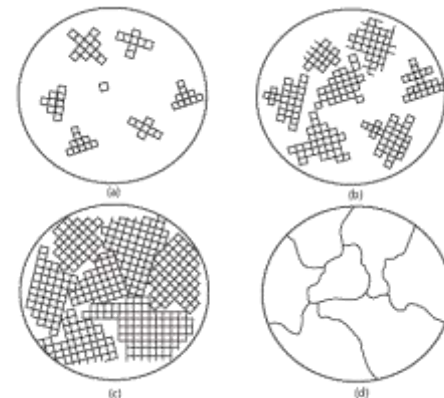
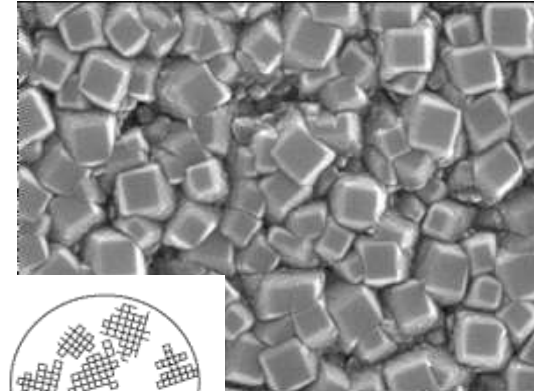
Single Crystal

- the periodic arrangement of atoms extends throughout the entire sample
- difficult to grow, environment must be tightly controlled
- anisotropic materials

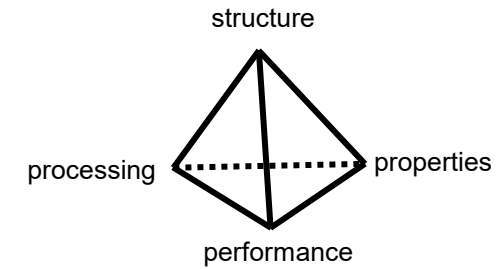


Polycrystalline

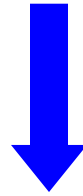
- many small crystals or grains
- small crystals misoriented with respect to one another
- several crystals are initiated and grow towards each other
- anisotropic or isotropic materials



Levels of Structure



STRUCTURE (length scale)



Sub-atomic

Atomic

< 0.2 nm

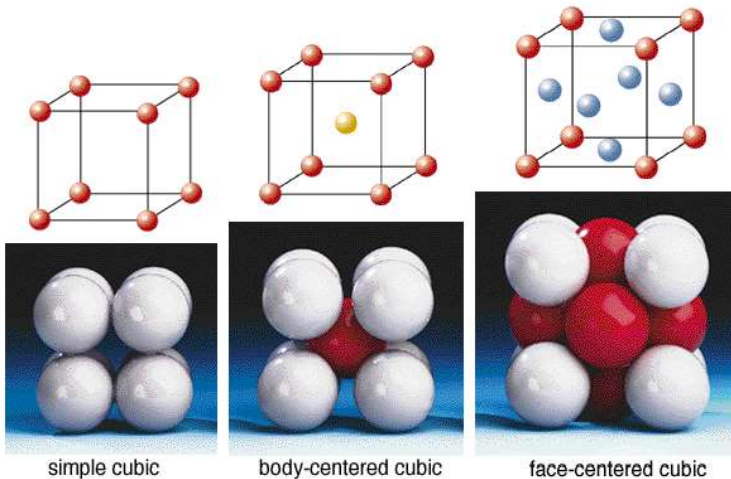
0.2-10 nm

1 nm = ?

Atomic Arrangement: Ordered vs. Disordered

Crystalline:

atoms are arranged in a 3D, periodic array giving the material “*long range order*”

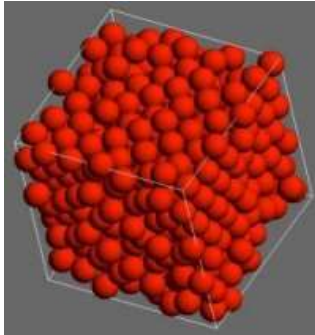


hexagonal close-packed

- stacking can effect properties (i.e. ductility)
- anisotropic materials

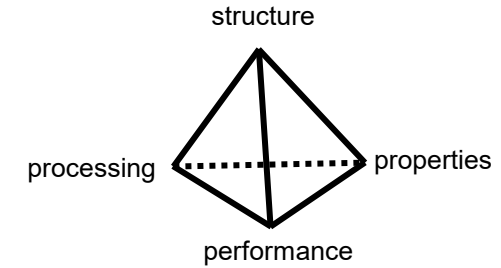
Non-crystalline or amorphous:

atoms only have short-range, *nearest neighbor order*



- viscous materials (generally complex formulas) or rapid cooling
- isotropic materials

Levels of Structure



STRUCTURE (length scale)



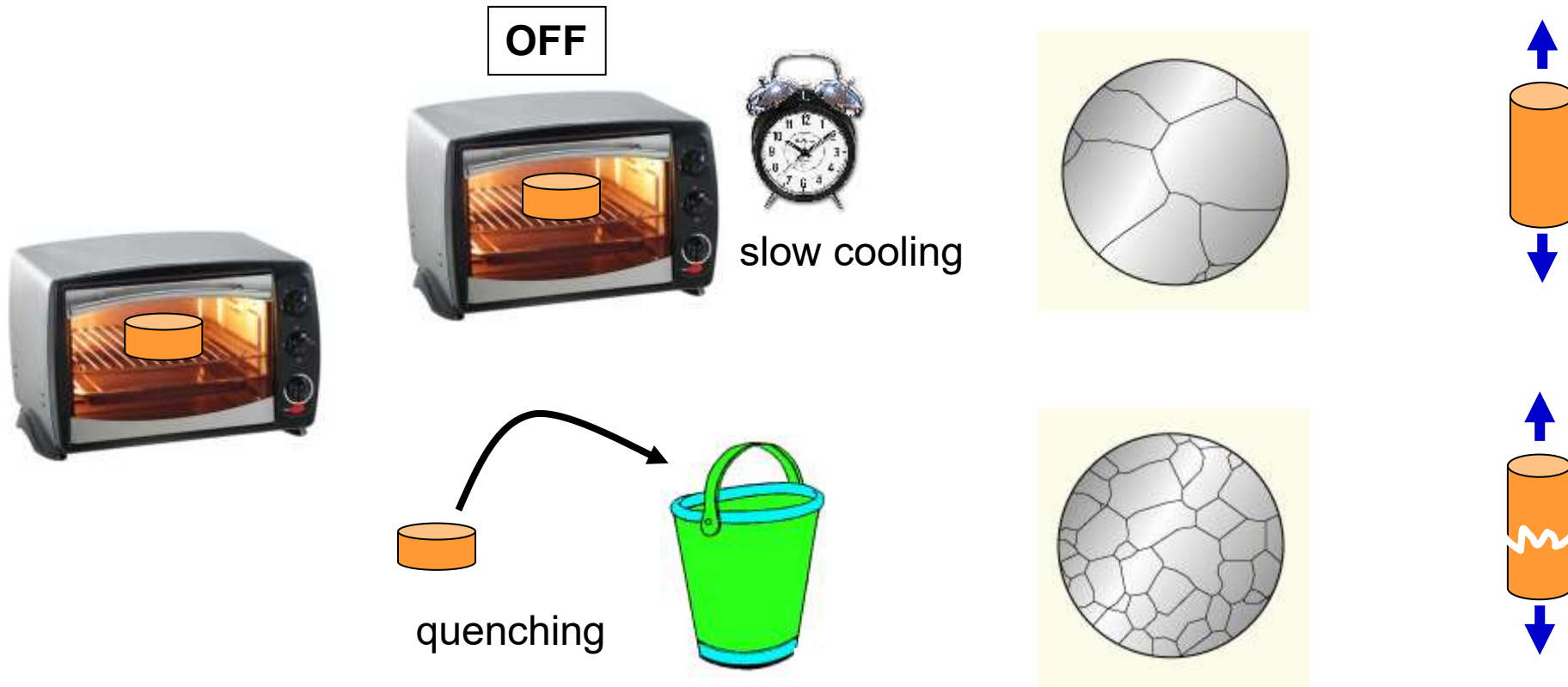
Sub-atomic

< 0.2 nm

Processing → Structure → Properties → Performance

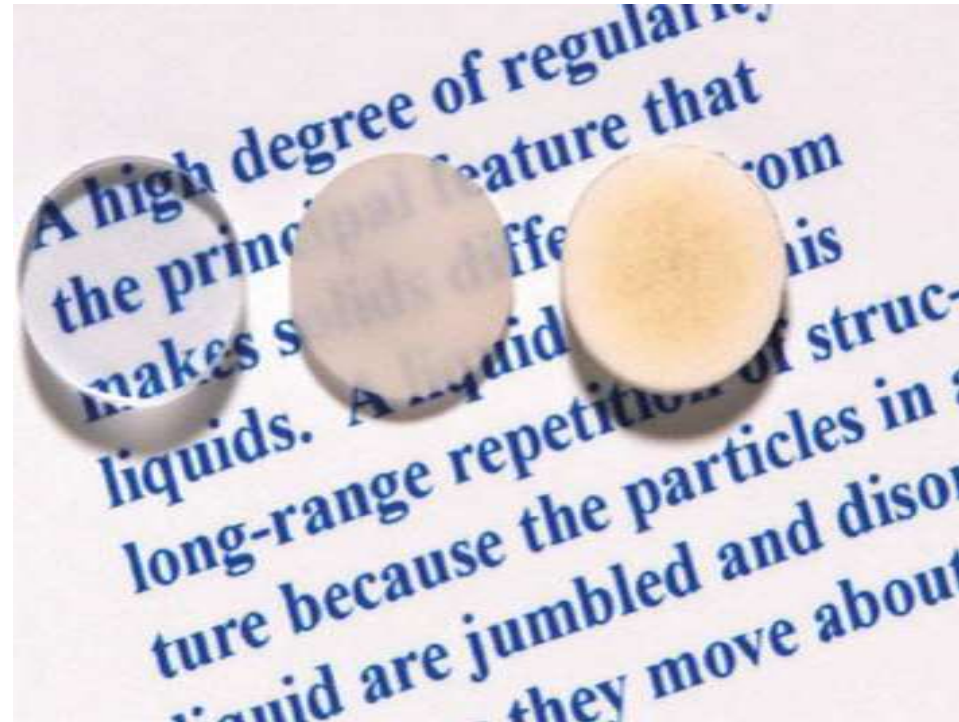
Performance Goal: increased strength from a metallic material

In actuality, crystals are NOT perfect. There are **defects**!
In metals, **strength** is determined by how easily defects can move!



Processing → Structure → Properties → Performance

Aluminum Oxide (Al_2O_3)

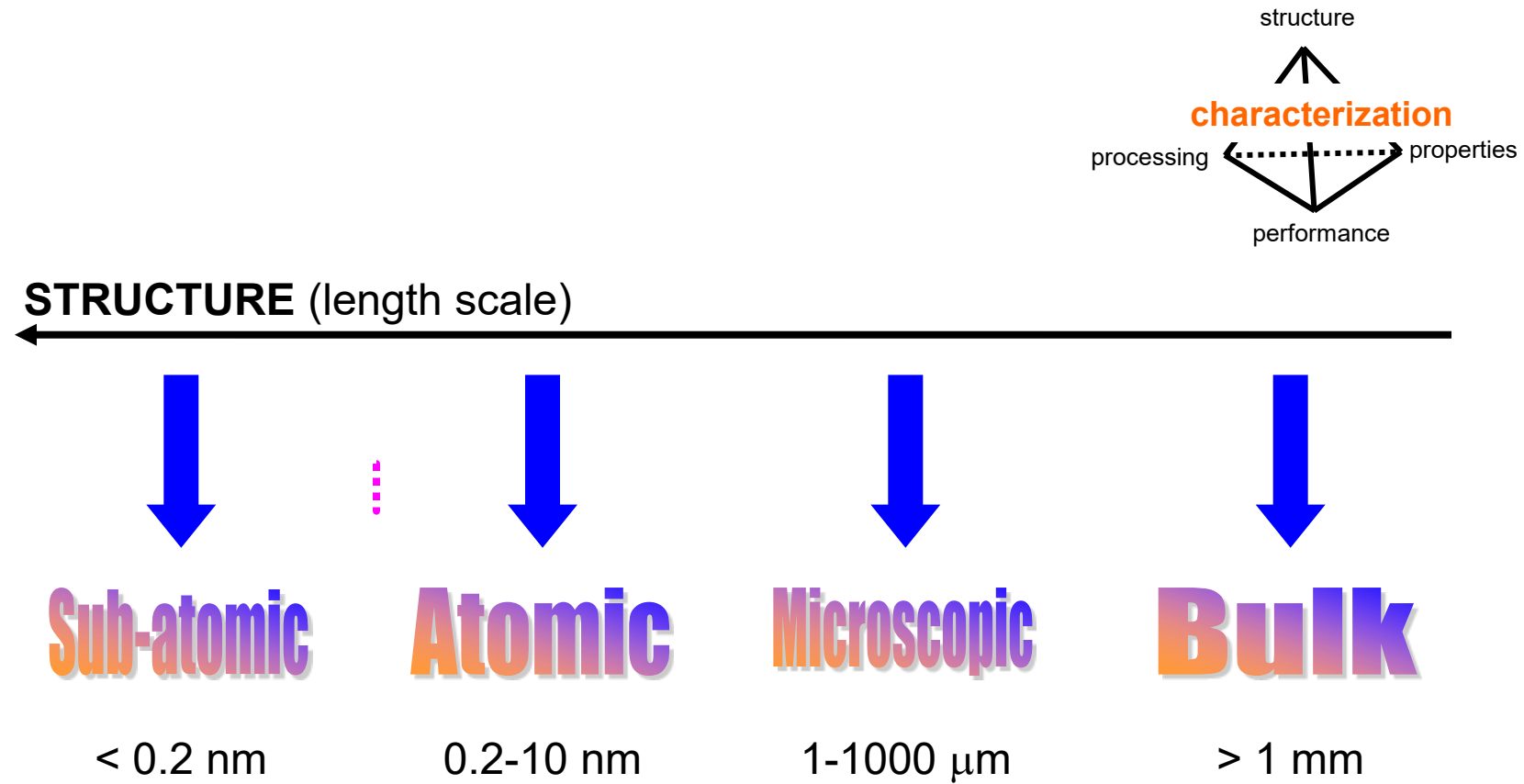


single-crystal
(transparent)

polycrystalline,
fully dense
(translucent)

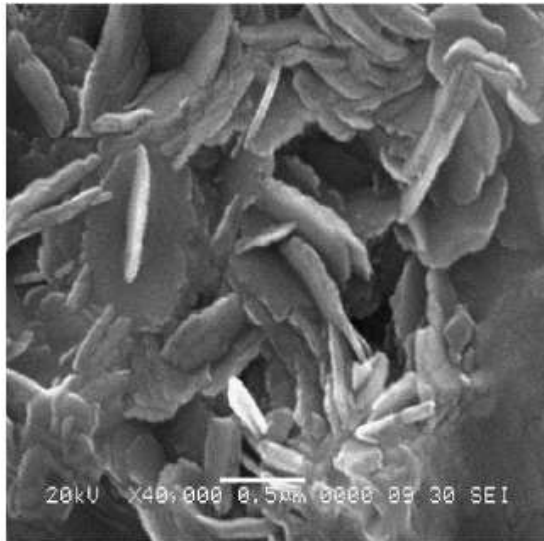
polycrystalline,
5% porosity
(opaque)

Characterization Techniques



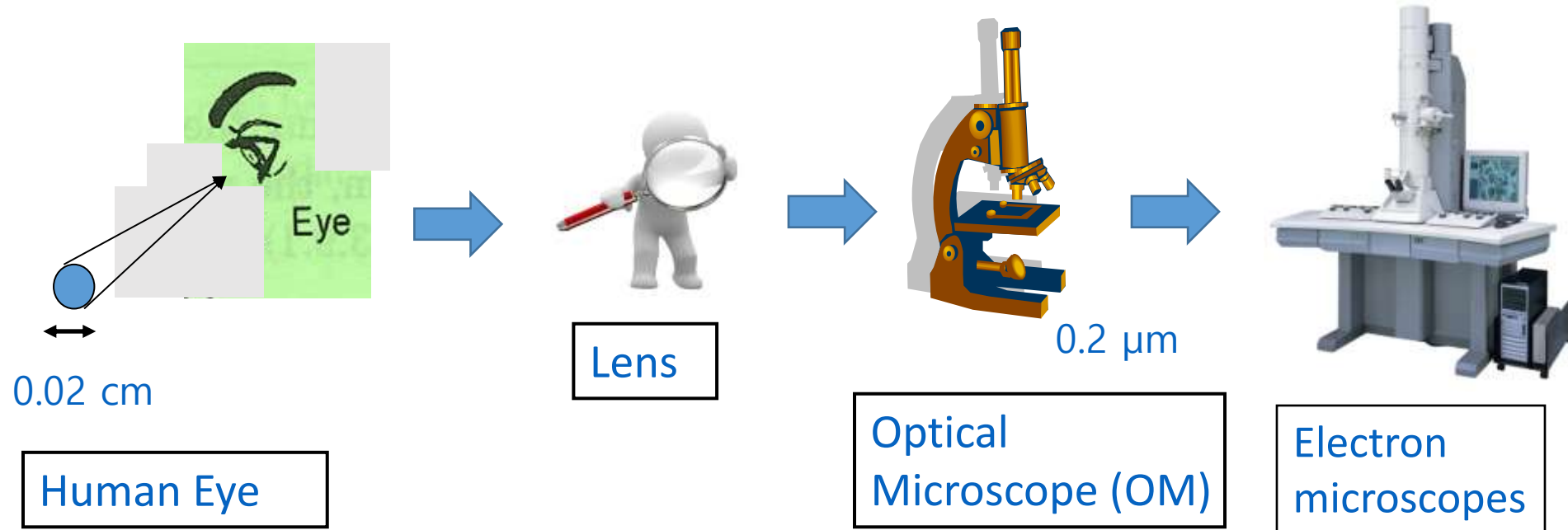
Morphology?

- How the surface looks like at micrometer/nanometer scale?
- Because surface of the sample plays very vital role in deciding its applicability in device fabrication.
- For example:
- (1) Gas Sensor: Porous and Nanocrystalline.
- (2) Solar cell: Smooth, Flat, Single/Microcrystalline.



A. Ghosh et al. Sensors and Actuators B 146 (2010) 69–74.

Why we need microscope?



The naked eye can see objects of dimensions 0.02 cm. To observe the particles of size smaller than this dimension, we need either huge lenses or an OM. However, the smallest size (d) of the object that can be observed by the OM has a limit of about 0.2 μm.

According to wave particle duality, a particle like electron of mass ' m ' accelerated through a potential difference of ' V ' volts is associated with it matter waves of wavelength

$$d = \frac{\lambda}{2(N.A.)}$$

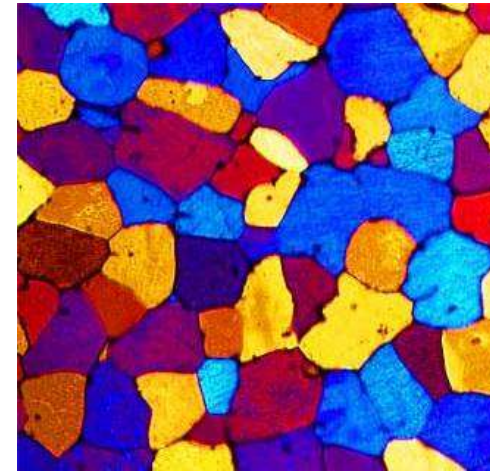
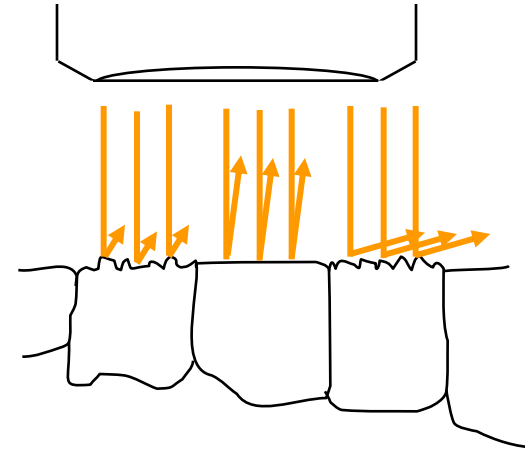
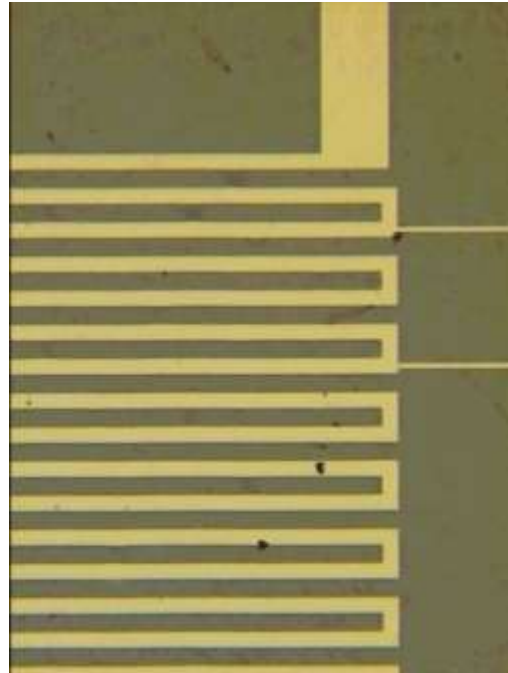
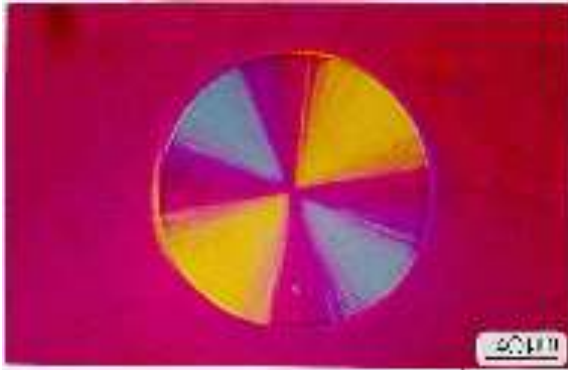
OM has the limit to about 500 X or 1000 X, because the smallest wavelength of visible light is of the order of 4000 Å.

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2meV}}$$

If $V = 60$ kV, the associated wavelength is 10^5 times smaller than the wavelength of visible light (0.4 μm). Hence the resolving power of electron microscope will be 10^5 times more than that of an OM.

Optical Microscopy

- light is used to study the microstructure
- opaque materials use reflected light, where as transparent materials can use reflected or transmitted light

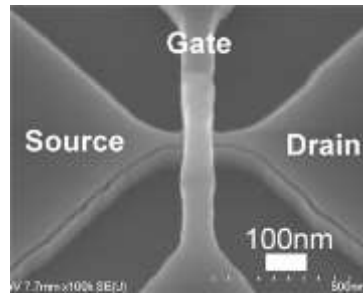
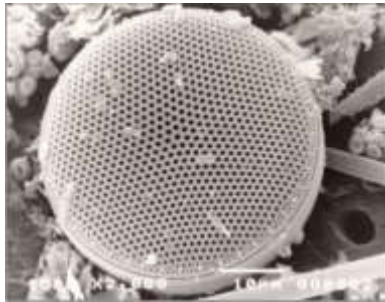


Electron Microscopy

- beams of electrons are used for imaging
- electrons are accelerated across large voltages
- a high velocity electron has a wavelength of about 0.003 nm
- the electron beam is focused and images are formed using magnetic lenses
- reflection and transmission imaging are both possible

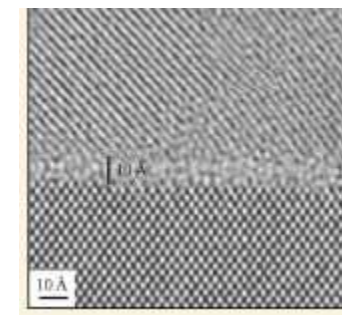
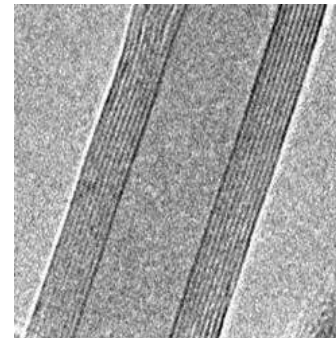
Scanning Electron Microscopy (SEM)

- an electron beam scans the surface and the reflected (backscattered) electrons are collected
- **sample must be electrically conductive**
- material surface is observed
- 200,000x magnification possible

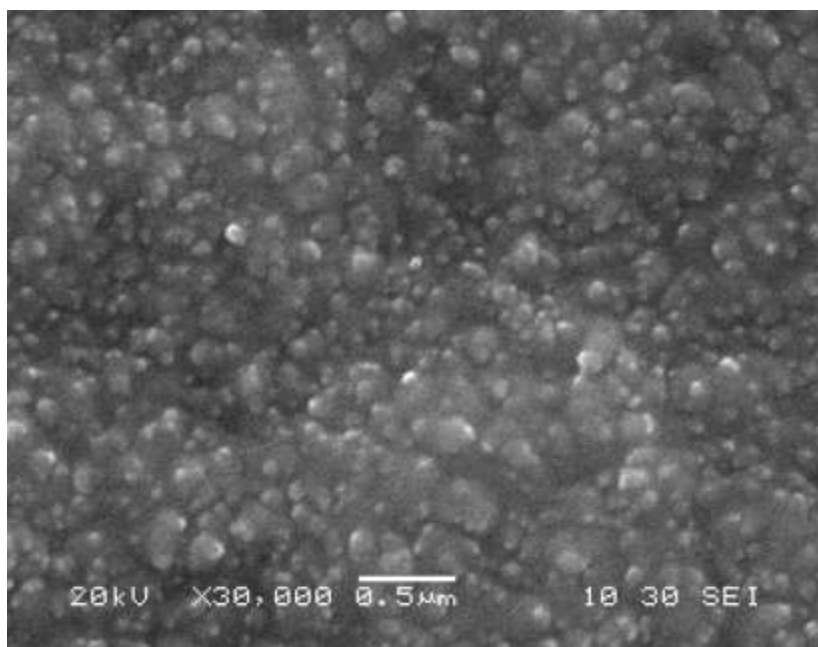
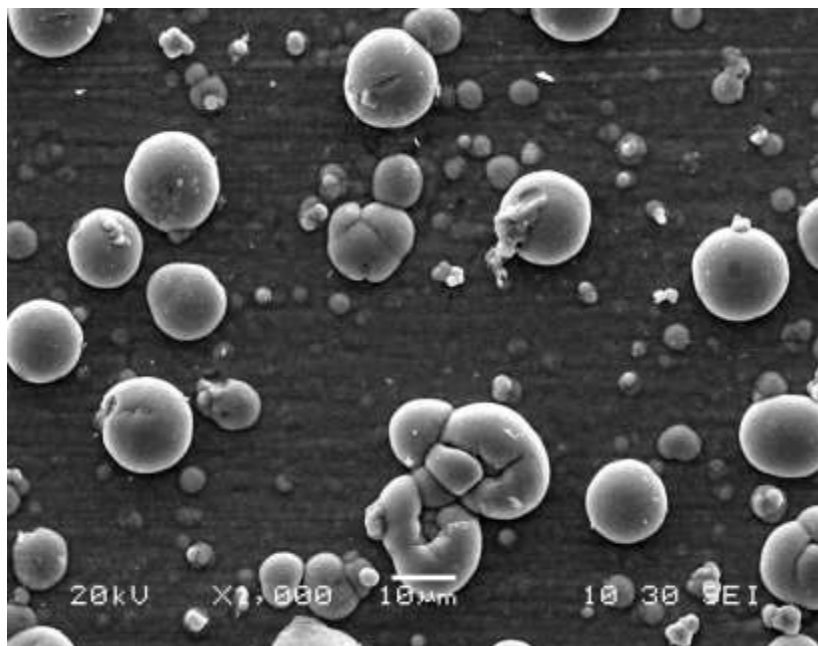


Transmission Electron Microscopy (TEM)

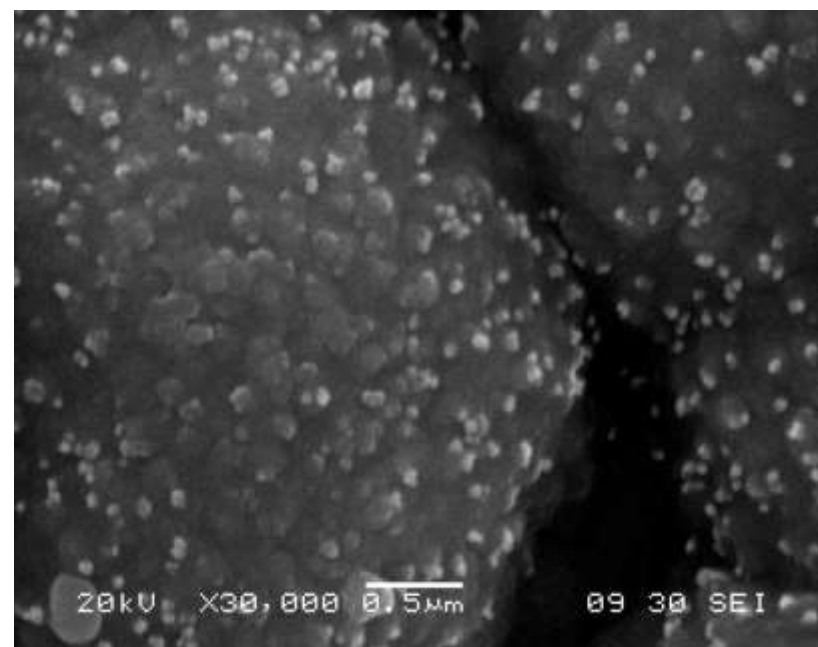
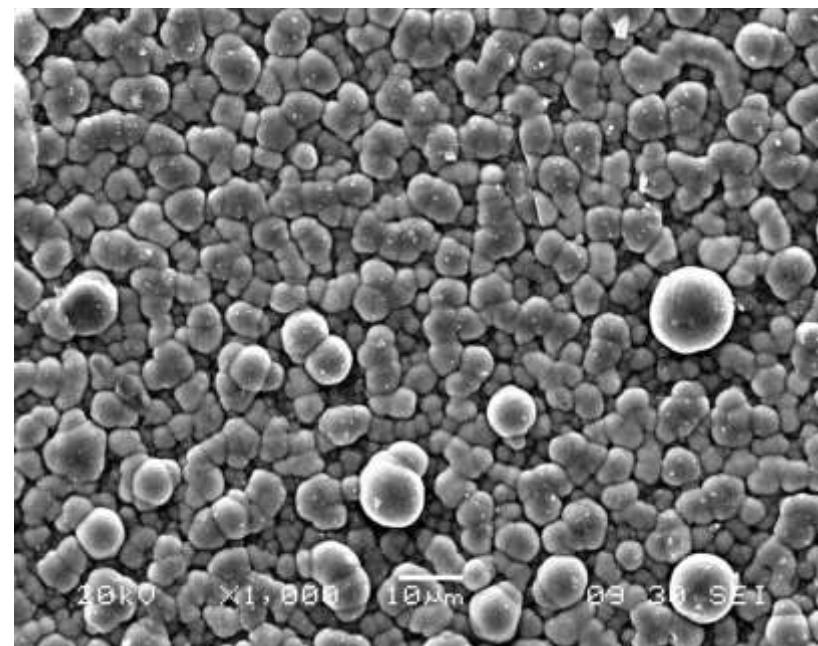
- an electron beam passes through the material
- thin samples
- details of internal microstructure observed
- 1,000,000x magnification possible



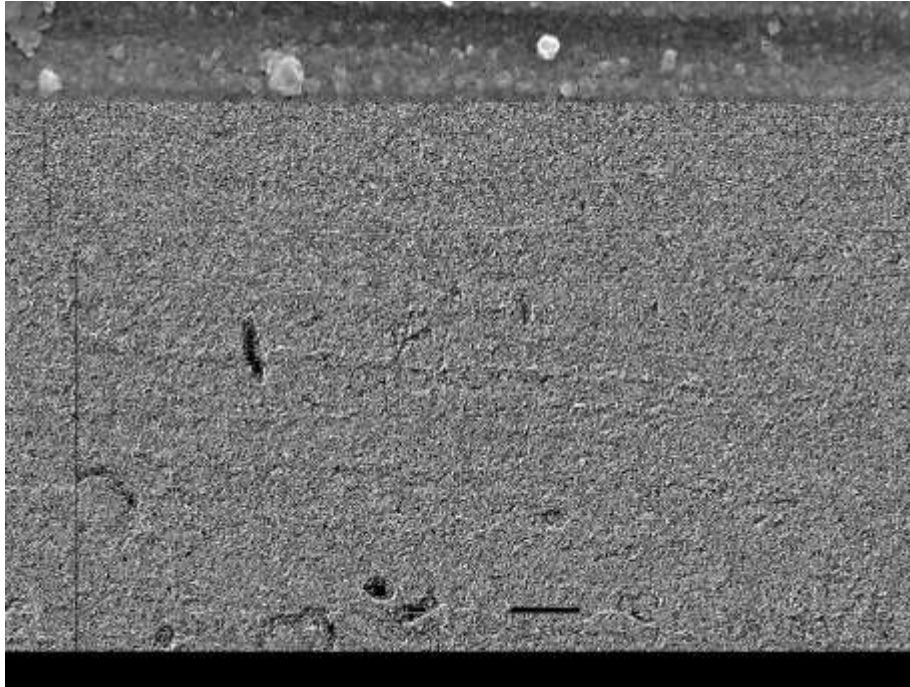
CdSe/SS



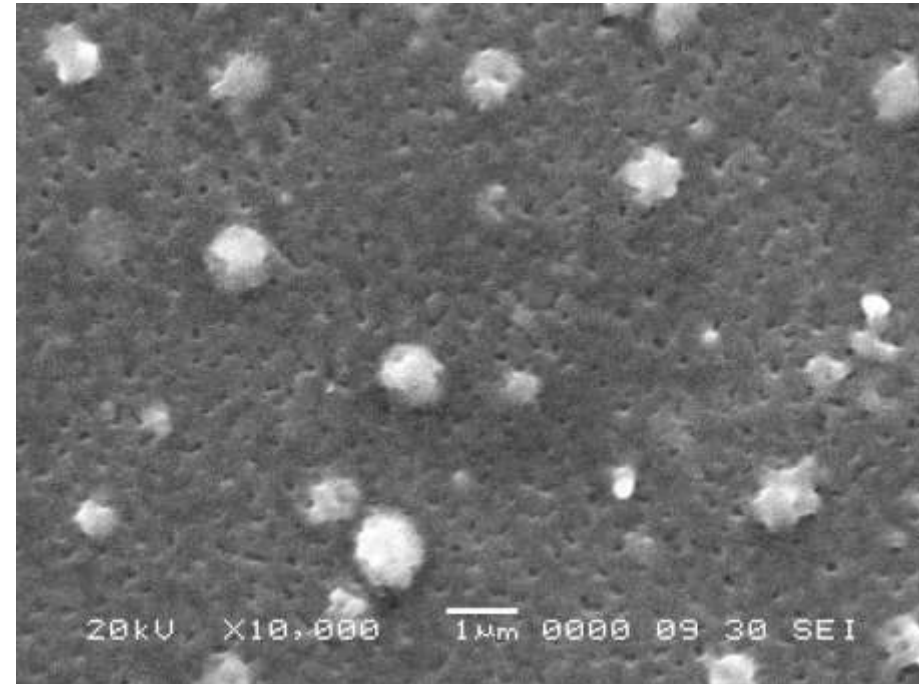
CdSe/FTO



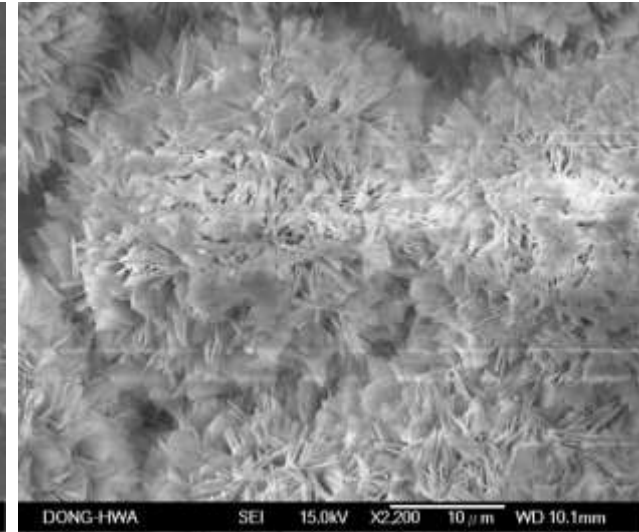
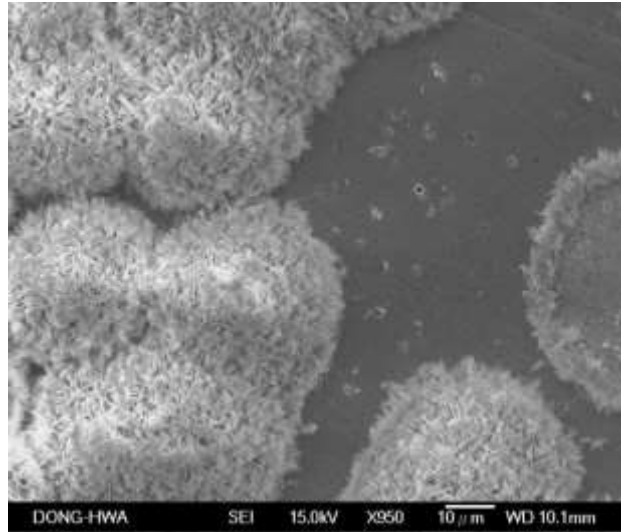
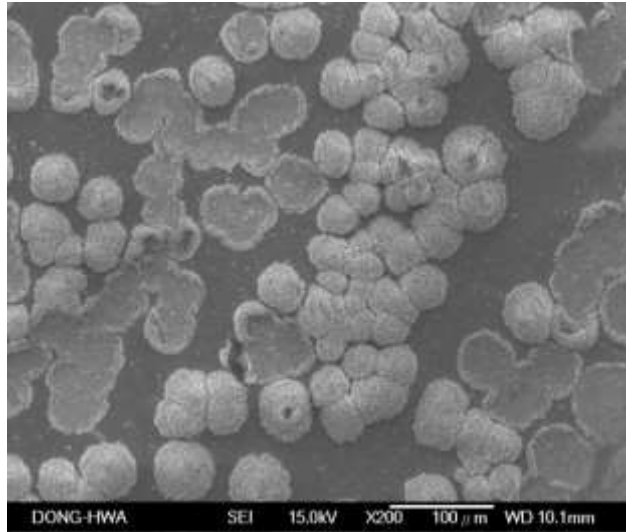
ZnSe/SS



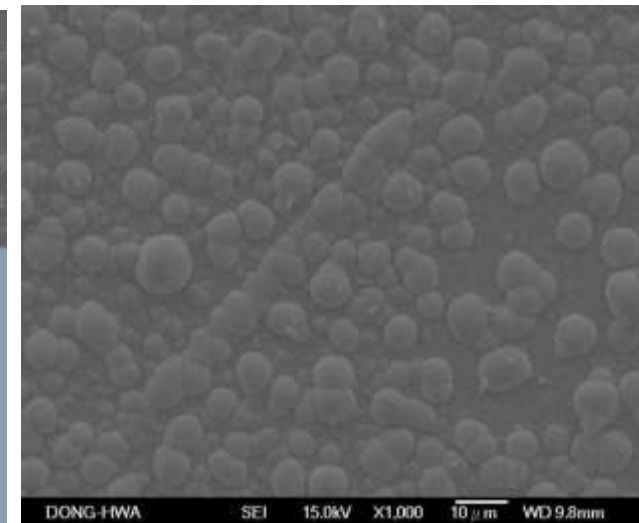
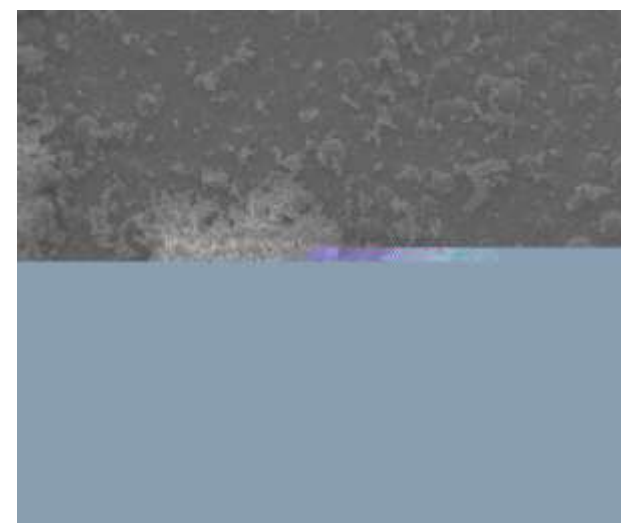
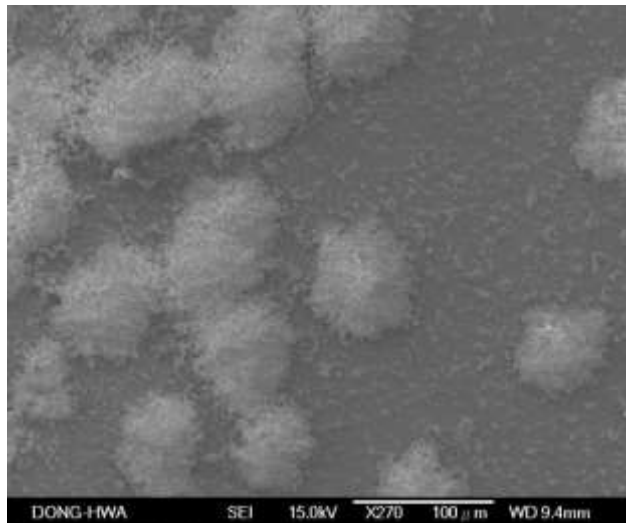
ZnSe/FTO



$\text{Cd}_{0.8}\text{Zn}_{0.2}\text{Se/SS}$



$\text{Cd}_{0.8}\text{Zn}_{0.2}\text{Se/FTO}$



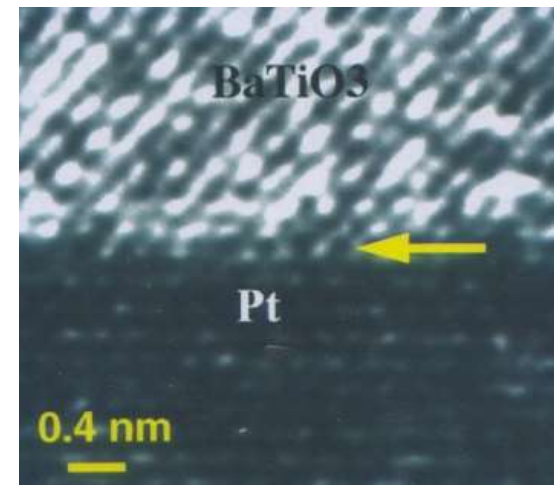
More detailed morphological study is done by TEM

The uniqueness of TEM is the ability to obtain:

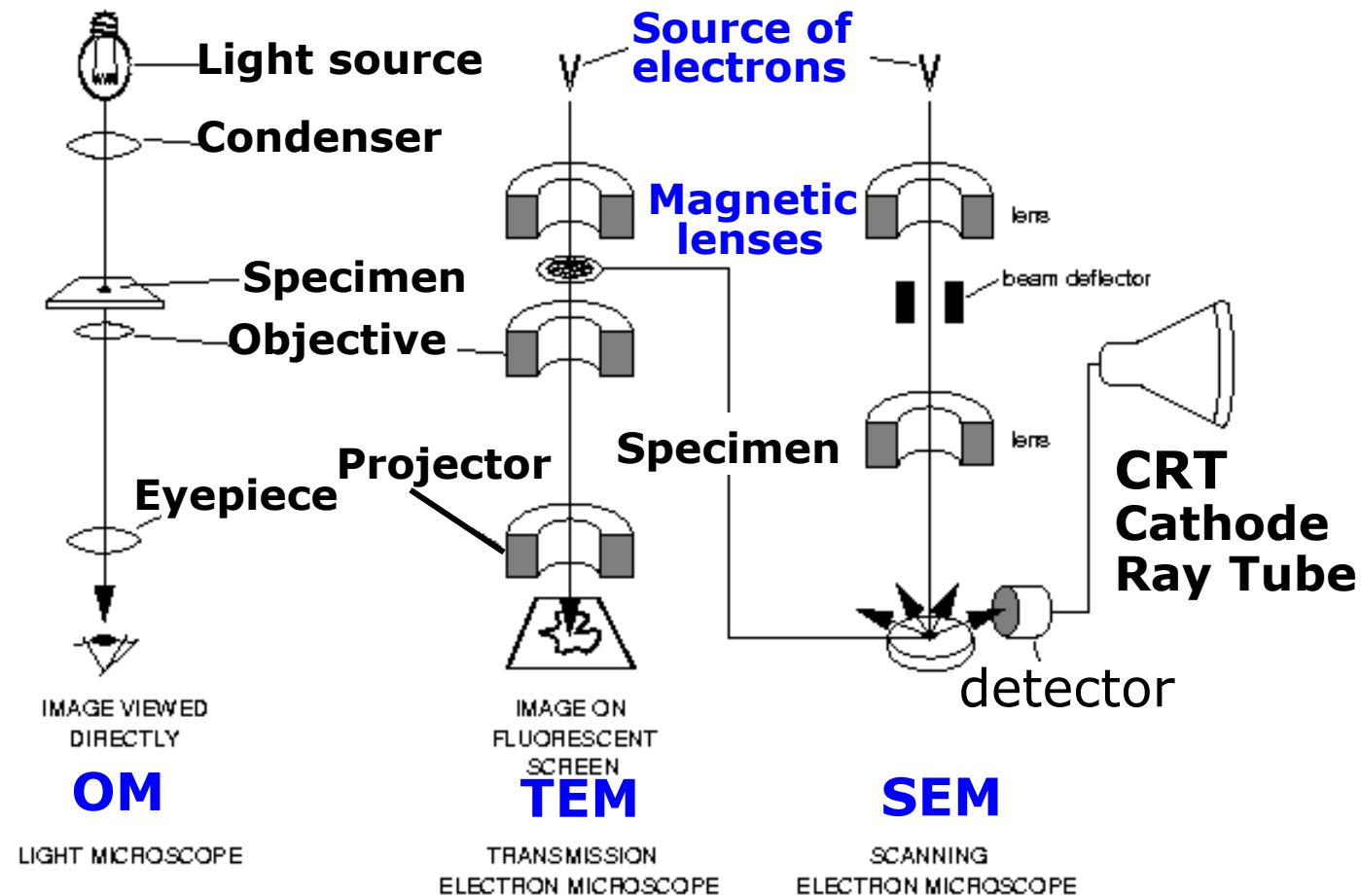
- 1) **morphological** (lattice constant, crystallite size, grain boundary and interface, secondary phase and distribution, defects and their nature, etc.),
- 2) **crystallographic**, atomic structural and micro-analytical such as chemical composition (at nm scale),
- 3) **bonding** (distance and angle),
- 4) **electronic** structure, coordination number data from the sample.

Limitation

More or less bulk-like information, the sample cannot be too thick, Sample preparation can be difficult.

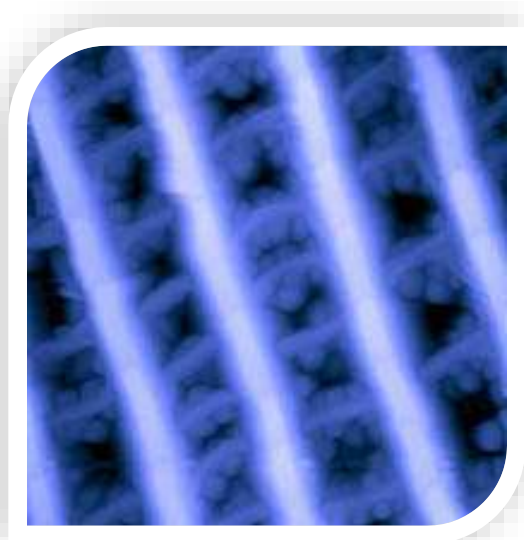


Comparison of OM, TEM and SEM

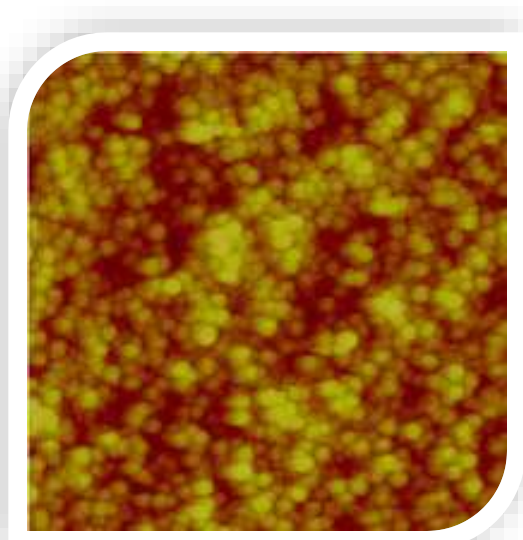


Principal features of an optical microscope, a transmission electron microscope and a scanning electron microscope, drawn to emphasize the similarities of overall design.

HIGH-RESOLUTION IMAGING WITH FORCES (ATOMIC FORCE MICROSCOPY)



SPM image of a butterfly wing.



*SPM image of silica coated
gold nanoparticles.*



SPM image of 70 nm photoresist lines.

History of a new microscopy

- The Scanning Tunneling Microscope (STM) was invented by **G. Binnig and H. Rohrer**, for which they were awarded the Nobel Prize in 1984.
- Atomic force microscopy (AFM) was developed when people tried to extend STM technique to investigate the electrically non-conductive materials, like proteins.
- In 1986, **Binnig and Quate** demonstrated for the first time the ideas of AFM, which used an ultra-small probe tip at the end of a cantilever (**Phys. Rev. Letters, 1986, Vol. 56, p 930**).
- In 1987, **Wickramsinghe** et al. developed an AFM setup with a vibrating cantilever technique (**J. Appl. Phys. 1987, Vol. 61, p 4723**), which used the light-lever mechanism.
- Currently AFM is the most common form of scanning probe microscopy

Introduction to Atomic Force Microscopy

What is AFM?

AFM is a type of scanning probe microscopy that provides high-resolution imaging and surface characterization at the nanoscale.

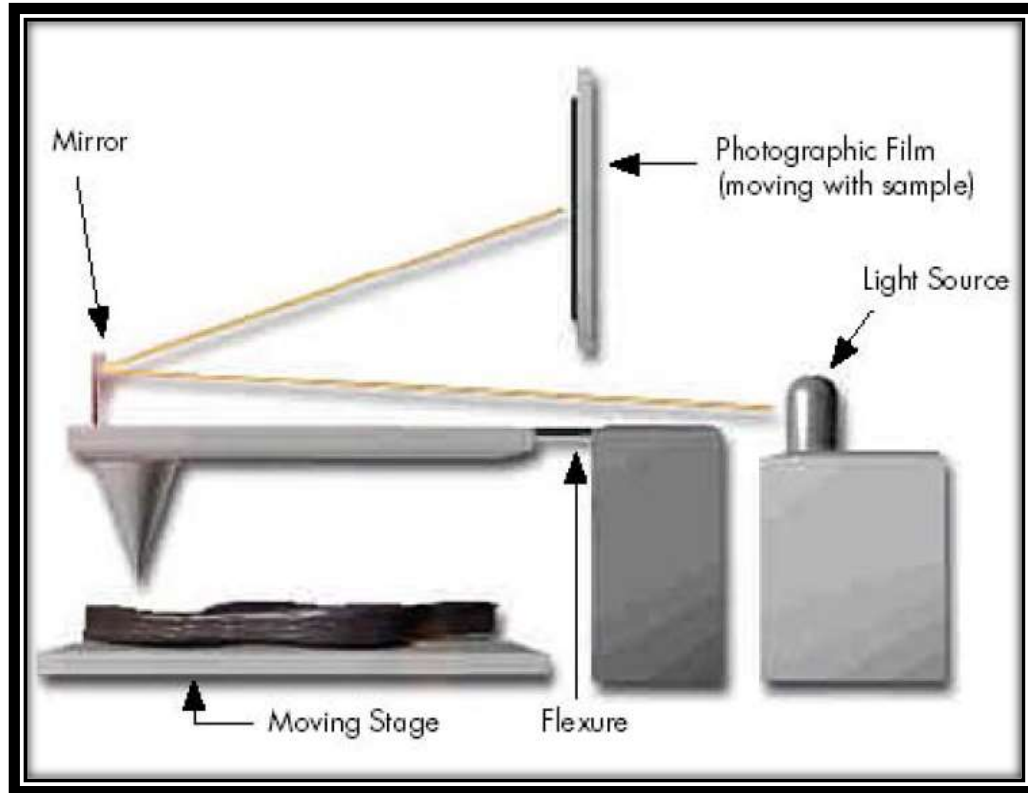
Purpose:

To study the surface topography and mechanical properties of materials at atomic or molecular scale.

Key Features:

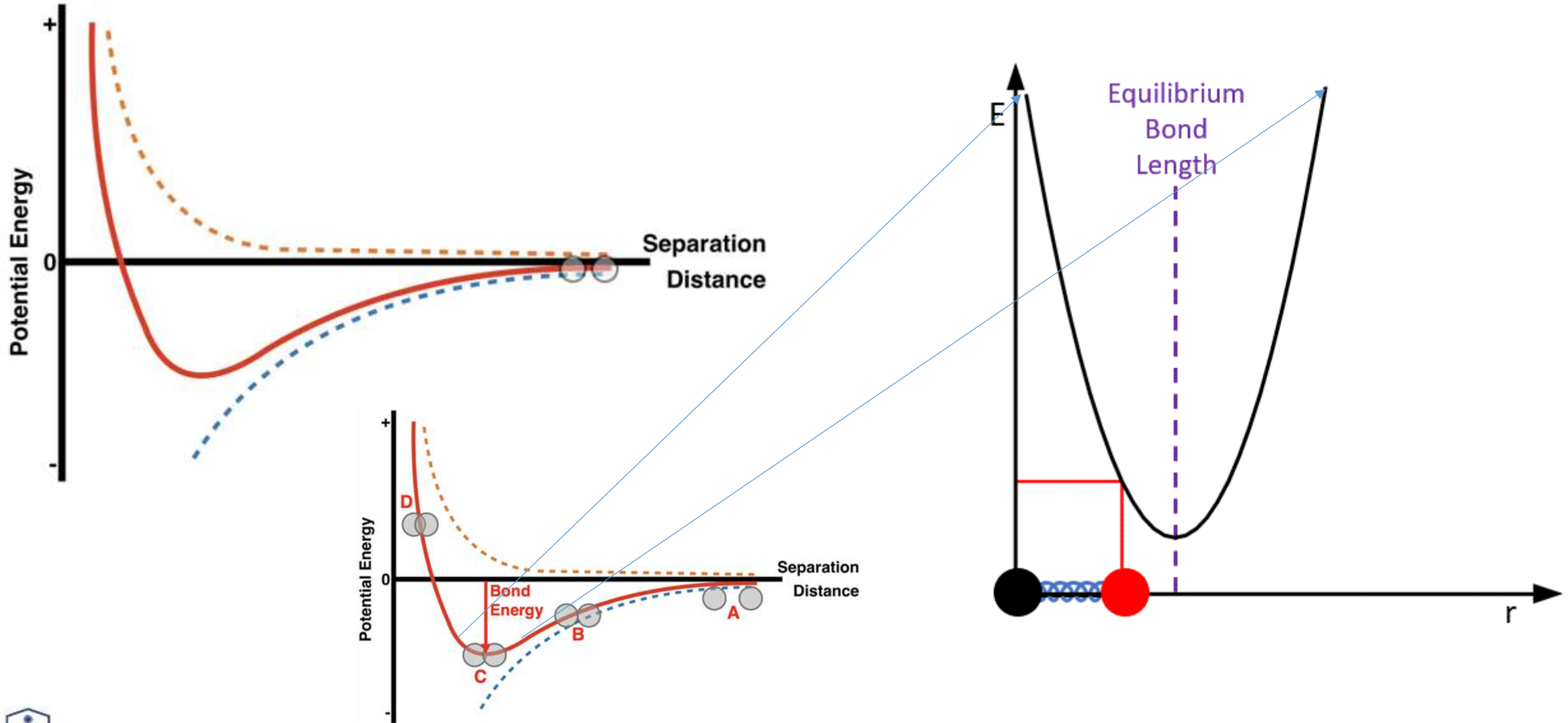
- ✓ High resolution (nanometer to sub-nanometer)
- ✓ Non-destructive
- ✓ Can be used in air, liquid, and vacuum environments

The surface profiler was invented by Schmalz in 1929.

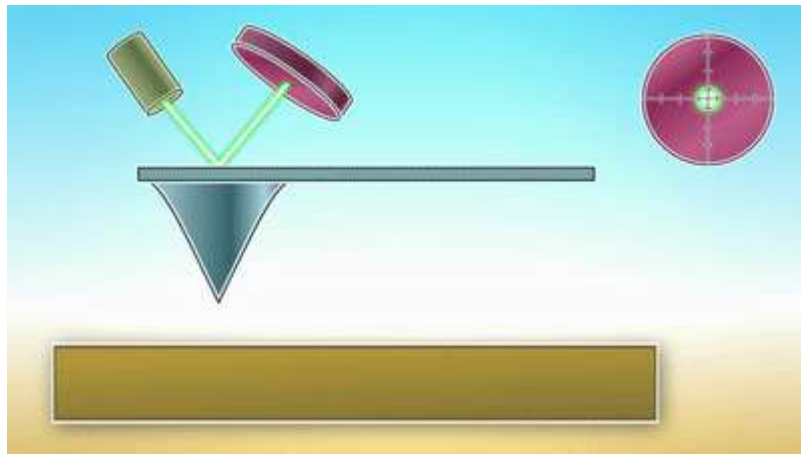


- ❖ It was designed to measure the surface roughness and texture of materials with high precision.
- ❖ The device uses mechanical means to scan the surface and detect variations in height.
- ❖ Primarily used in material science, manufacturing, and quality control to assess surface characteristics.
- ❖ Pioneered the field of surface metrology, enabling more accurate analysis of material surfaces, which is critical in various industries such as engineering, electronics, and metallurgy.
- ❖ This invention laid the foundation for more advanced surface measurement techniques used in modern technology.

Basic Idea behind AFM



Principle of Atomic Force Microscopy



- **Basic Concept:**

- AFM uses a sharp tip (probe) that scans the surface.
- The deflection of the tip is measured as it interacts with the sample surface.

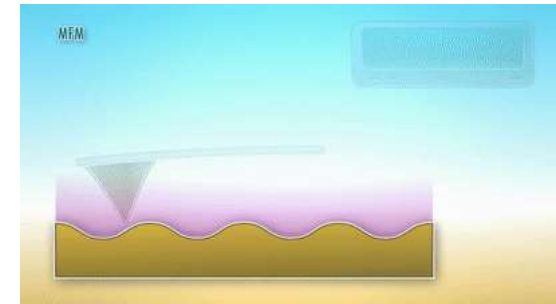
- **Working Mechanism:**

- A cantilever with a sharp tip is brought close to the sample surface.
- Forces between the tip and the sample cause the cantilever to bend.
- The deflection is measured using a laser beam reflected from the back of the cantilever.
- The deflection data is used to create an image of the surface.

AFM Imaging Modes

- **Contact Mode:**

- The tip maintains a constant contact with the surface, scanning it to measure topography.

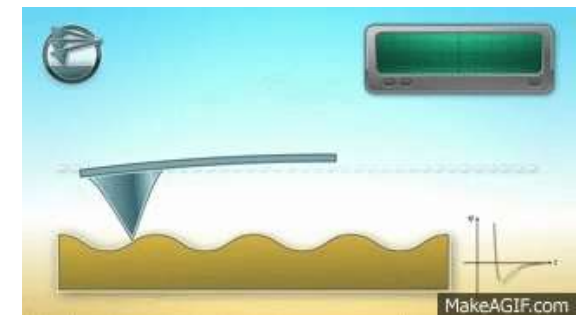


- **Non-contact Mode:**

- The tip oscillates above the surface without touching it, measuring attractive forces.

- **Tapping Mode:**

- The tip taps the surface intermittently, minimizing lateral forces.





Tips plays an important role...

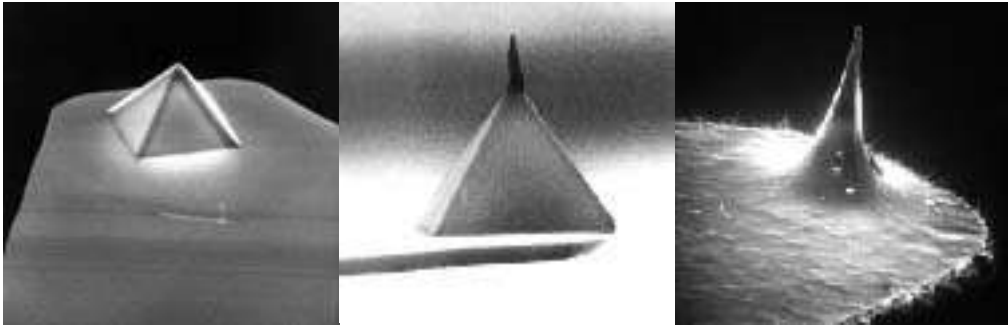
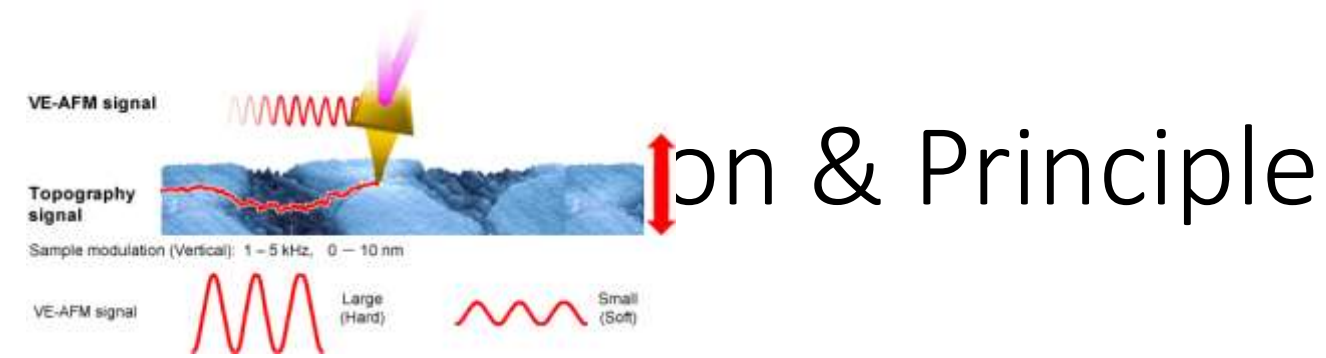


Figure 4. Three common types of AFM tip. (a) normal tip (3 μm tall); (b) supertip; (c) Ultralever (also 3 μm tall). Electron micrographs by Jean-Paul Revel, Caltech. Tips from Park Scientific Instruments; supertip made by Jean-Paul Revel.

<http://stm2.nrl.navy.mil/how-afm/how-afm.html#imaging%20modes>

Nanoindentation
of a virus
using
Atomic Force Microscopy

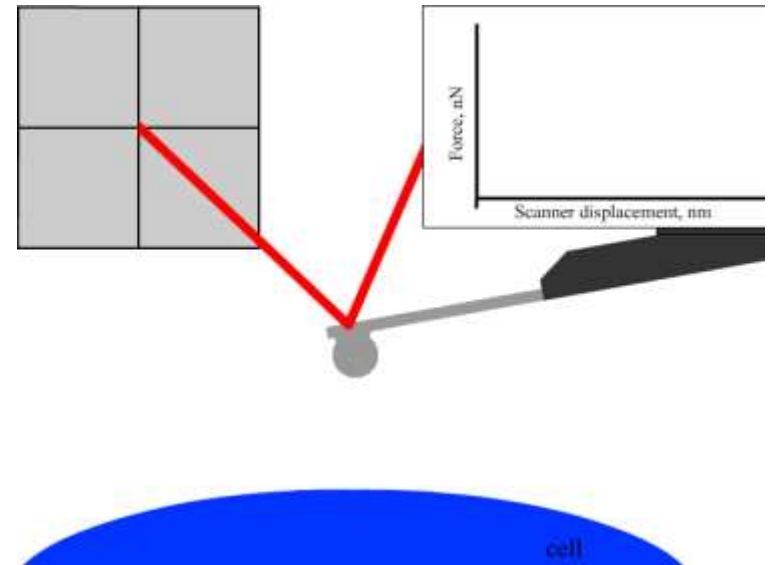
Wouter Roos
Vrije Universiteit
Amsterdam



- Nanoindentation Lithography (NIL) is a resist-free, direct-write lithography method.
- Uses a sharp indenter tip (AFM-like, diamond) to create nanoscale indentations.
- Resolution typically ranges from 10–100 nm.
- Controlled by indentation force, tip radius, and substrate properties.
- Enables nanoscale patterning without complex photomasks.

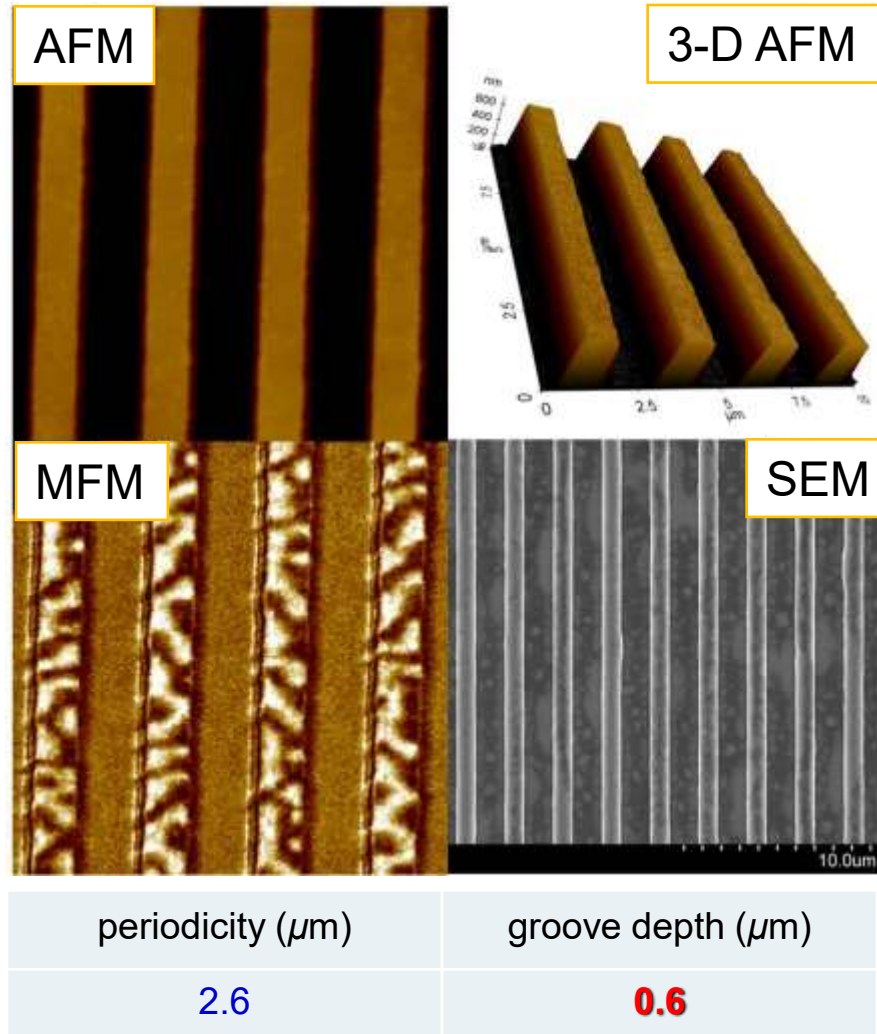
Process Flow of NIL

1. Substrate preparation (Si, glass, polymer, thin films).
2. Tip indentation with controlled load.
3. Pattern formation: nanoscale pits or lines.
4. Pattern transfer (optional):
 - Reactive Ion Etching (RIE)
 - Thin film deposition
 - Nanoparticle assembly



1D-Ni spin photonic crystals (SPCs)

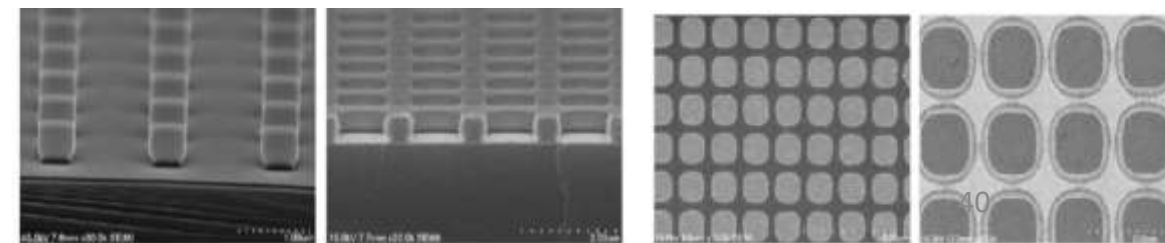
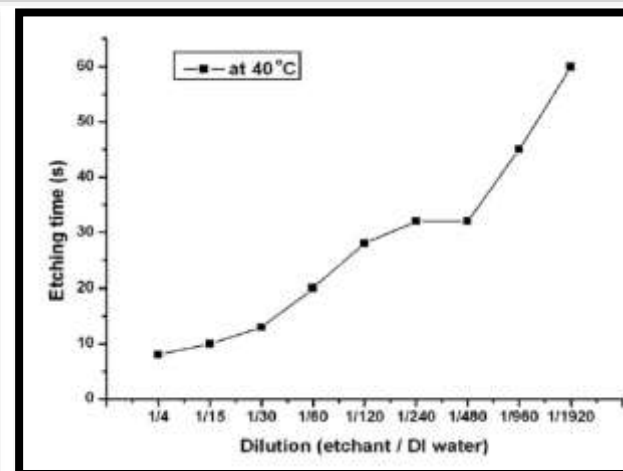
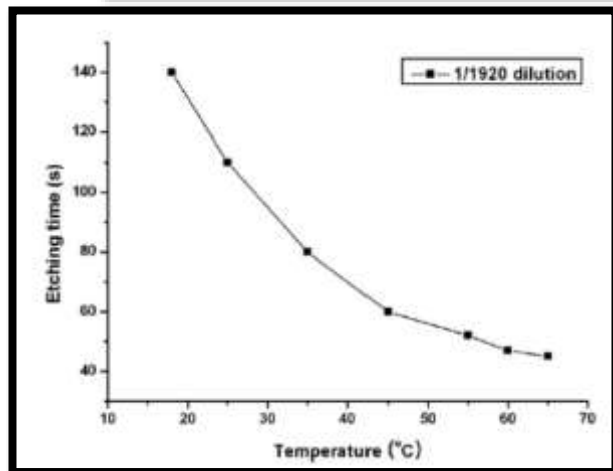
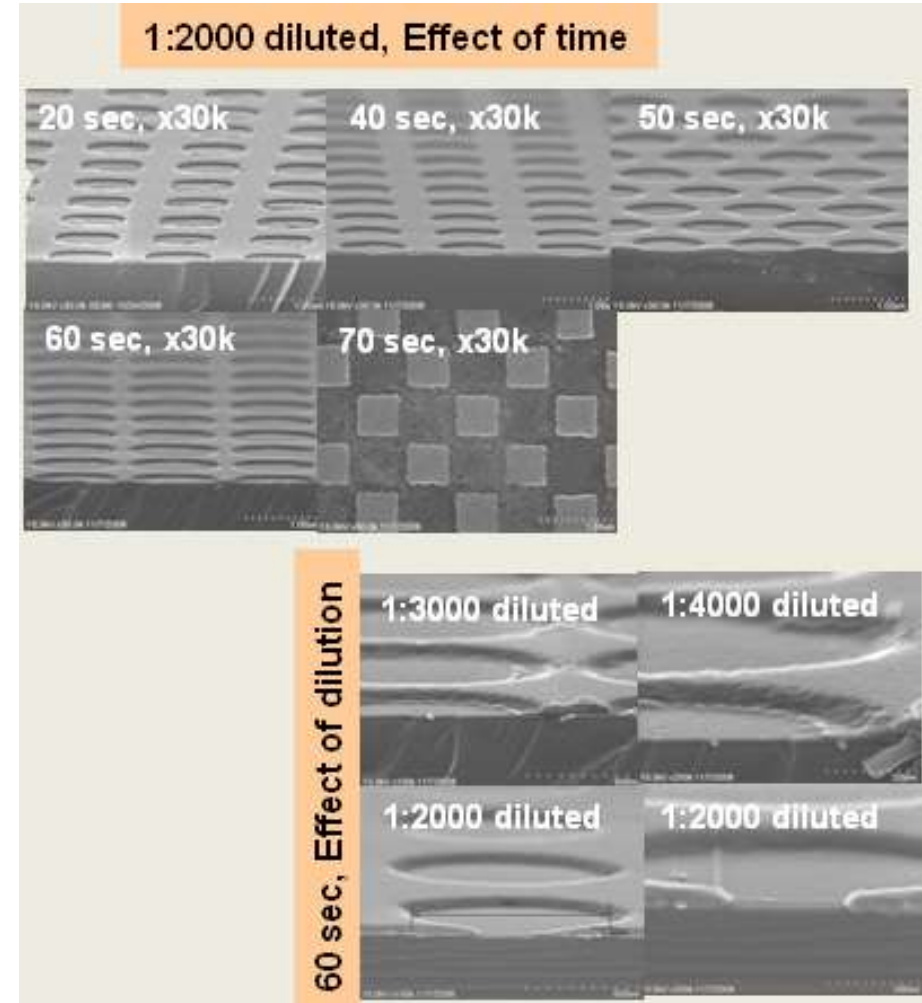
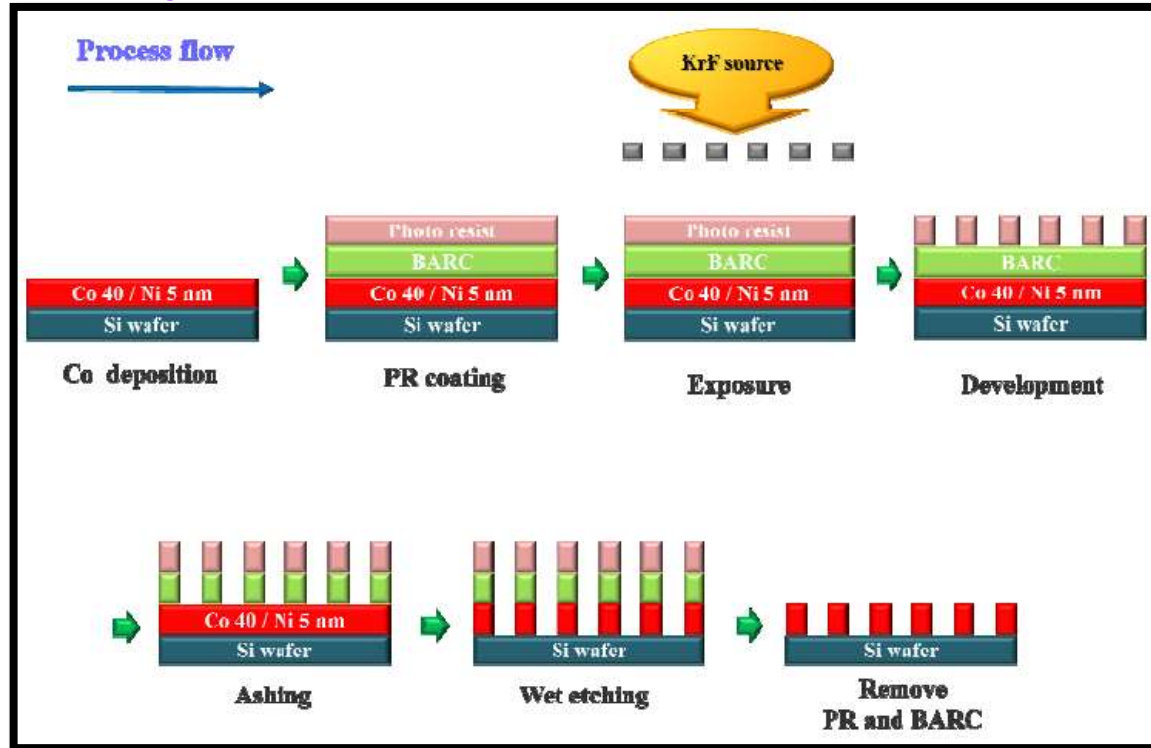
Surface morphology & magnetic domain structures



- By comparing the AFM and the MFM images, it is clear that the topography and the magnetic contrast are well separated.
- The domain structure within the ferromagnetic stripes in the demagnetized state can be described as simple domains (single spin direction).
- It is well known that the domain walls are not only perpendicular or parallel to the stripes but make intermediate angles in ferromagnetic Ni grating system.

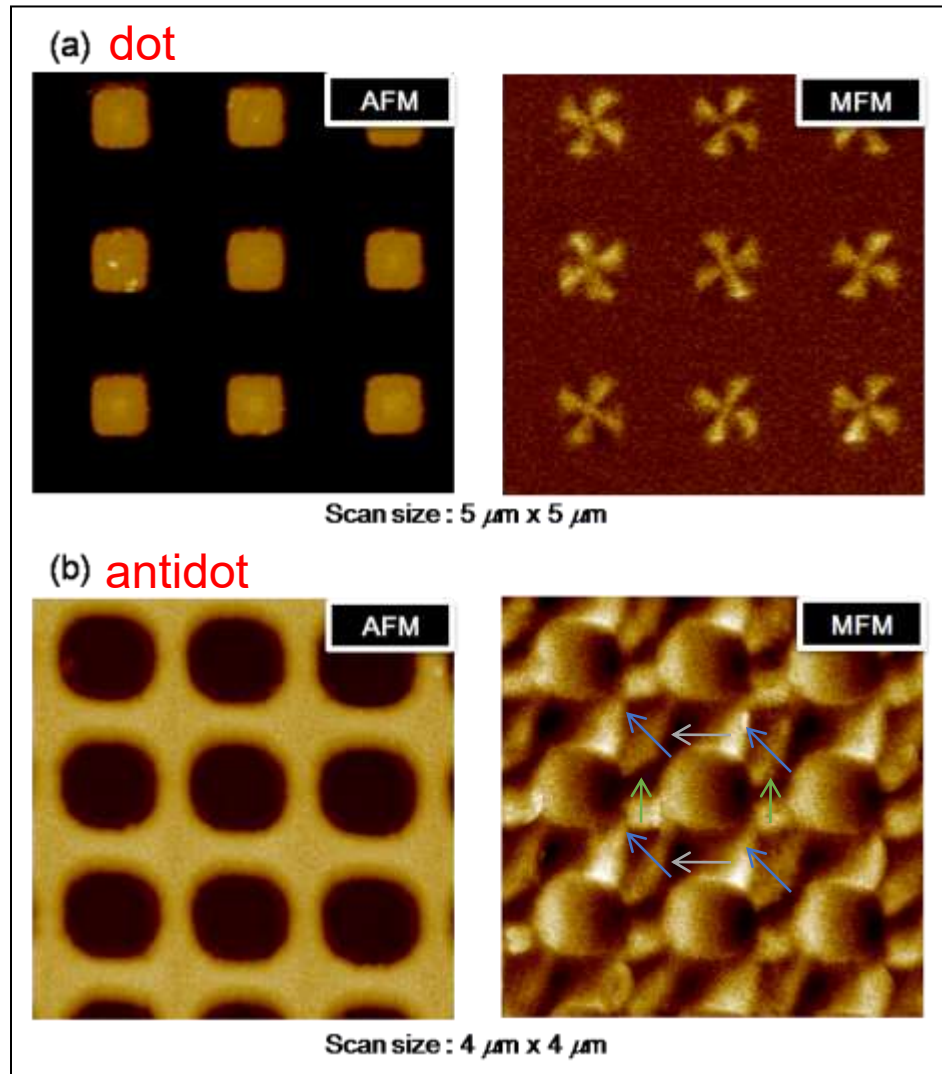
PHOTOLITHOGRAPHY & WET ETCHING

Patterning was done in Korea Advanced Nano Fab center (KANC), Suwon in a clean room



2D-spin photonic crystals (SPCs)

Direct observation of change of magnetic domains in patterned arrays of cobalt magnetic nanostructures



- The MFM image of the square pattern shown in Fig. (a) clearly demonstrates a Landau–Lifshitz-type flux closure domain structure with 90°-domain walls and a central vortex.
 - ✓ The balance of demagnetization and exchange energies results in flux-closure structures such as the fourfold -closure-domain pattern in thin squares.
- The MFM image in Fig. (b) has three different domains.
 - ✓ Two are wedge-shaped domains connecting the holes at each end and one is a diamond-shaped domain.
 - ✓ This configuration minimizes magnetostatic energy

1. What is characterized using Surface Characterization?

Surfaces may be characterized with respect to their topography (i.e. roughness), chemistry, surface orientation, and thickness of chemically homogeneous regions at the surface. Typically, mean free path of probe into sample is low or sample is comprised only of surface atoms.

2. What are the major categories of Surface Measurement Techniques?

- Contact Methods vs. Non-contact Methods

3. Which Surface Characterization Techniques will we learn about in this lecture?

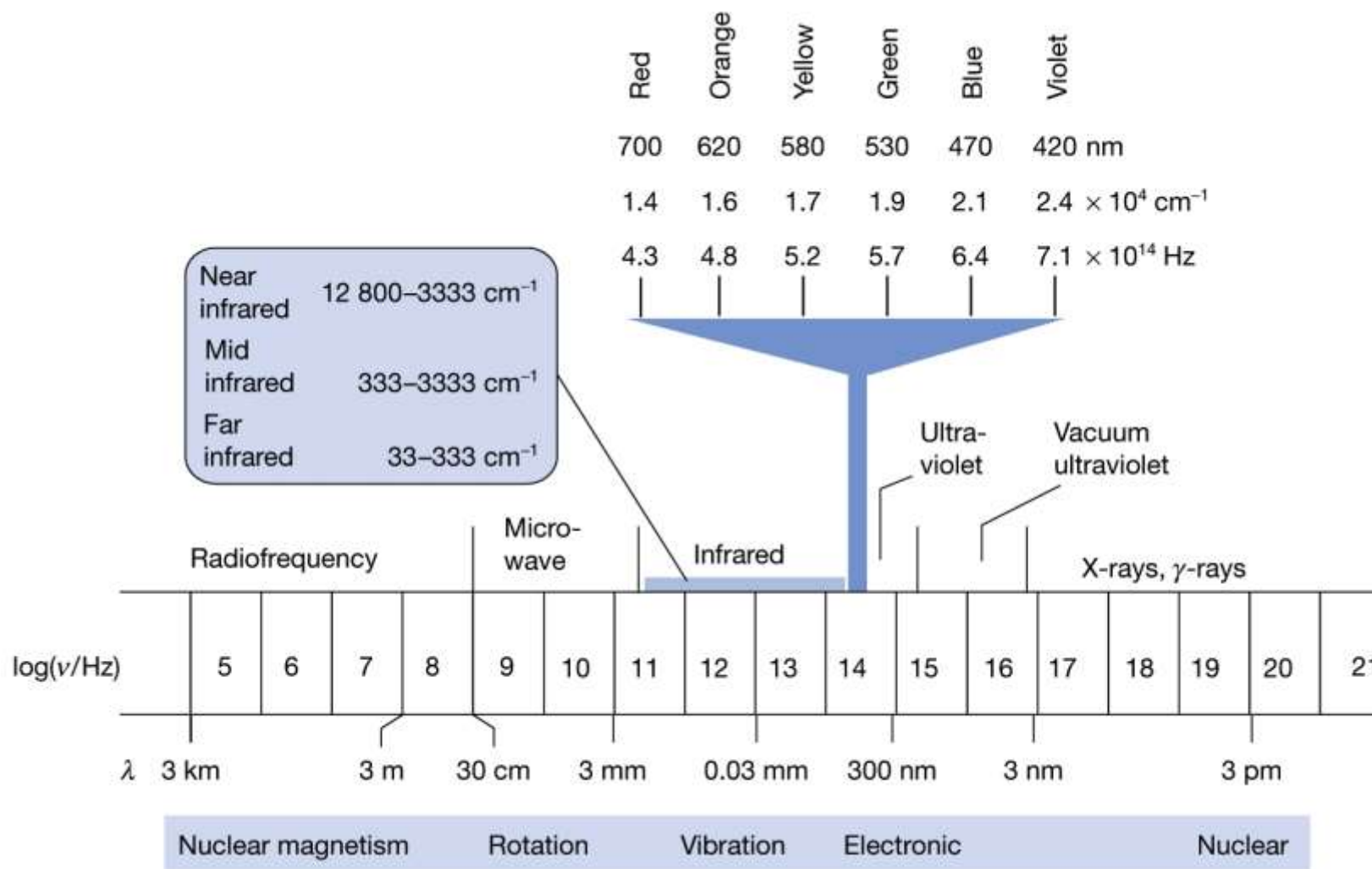
- X-ray diffraction (XRD)
- Energy Dispersive analysis by X-rays (EDAX)
- Electron microscopy (EM): SEM and TEM
- Atomic Force Microscopy (AFM)
- UV-VIS Spectroscopy
- Near-IR Spectroscopy
- Raman Spectroscopy
- X-ray Photoelectron Spectroscopy (XPS) also called Electron Spectroscopy for Chemical Analysis (ESCA)

4. What are some other Surface Characterization Techniques of practical importance in research?

- Rutherford Backscattering Spectroscopy
- Low Energy Ion Scattering Spectroscopy
- TOF-Secondary Ion Mass Spectrometry (TOF-SIMS)
- Auger Electron Microscopy (AES)
- Low Energy Electron Diffraction (LEED)

Spectroscopy (UV-VIS, IR, RAMAN)

THE ELECTROMAGNETIC SPECTRUM



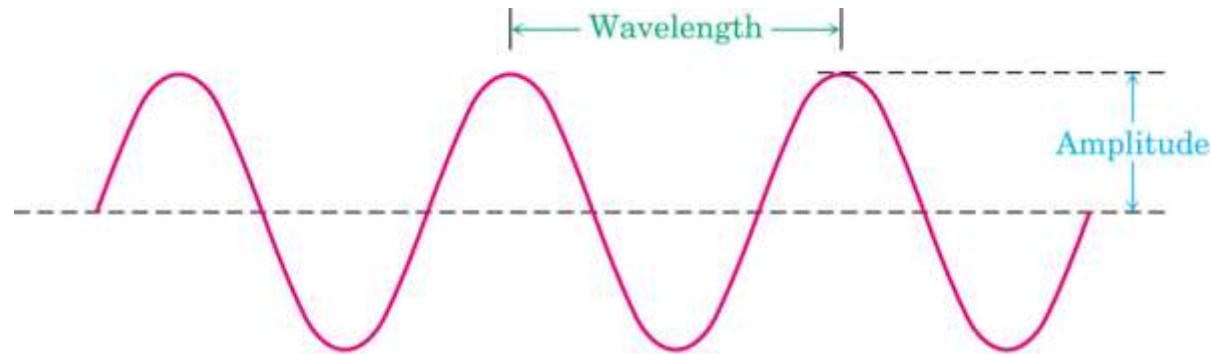
WHAT IS SPECTROSCOPY?

- Atoms and molecules interact with electromagnetic radiation (EMR) in a wide variety of ways.
- Atoms and molecules may absorb and/or emit EMR.
- Absorption of EMR stimulates different types of motion in atoms and/or molecules.
- The patterns of absorption (wavelengths absorbed and to what extent) and/or emission (wavelengths emitted and their respective intensities) are called '*spectra*'.
- The field of **spectroscopy** is concerned with the interpretation of *spectra* in terms of atomic and molecular structure (and environment).

Principles of spectroscopy: Electromagnetic radiation



Electromagnetic radiation has the properties of a particle (photon) *and* a wave.



λ = distance of one wave

ν = frequency: waves per unit time (sec^{-1} , Hz)

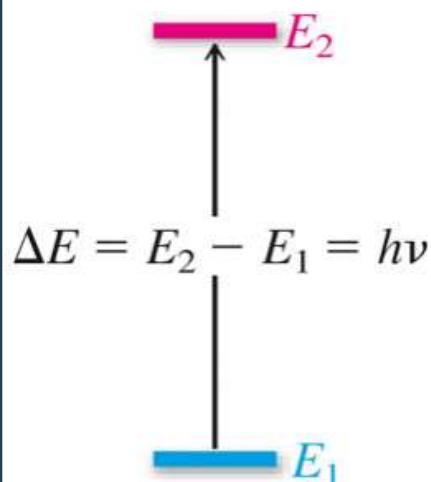
c = speed of light ($3.0 \times 10^8 \text{ m} \cdot \text{sec}^{-1}$)

h = Planck's constant ($6.63 \times 10^{-34} \text{ J} \cdot \text{sec}$)

Principles of molecular spectroscopy: Quantized Energy Levels

molecules have discrete energy levels

(no continuum between levels)



A molecule absorbs electromagnetic radiation when the energy of photon corresponds to the difference in energy between two states

UV-Vis: valence electron transitions

- gives information about π -bonds and conjugated systems

Infrared: molecular vibrations (stretches, bends)

- identify functional groups

Radiowaves: nuclear spin in a magnetic field (NMR)

- gives a map of the H and C framework

UV-Visible Spectroscopy

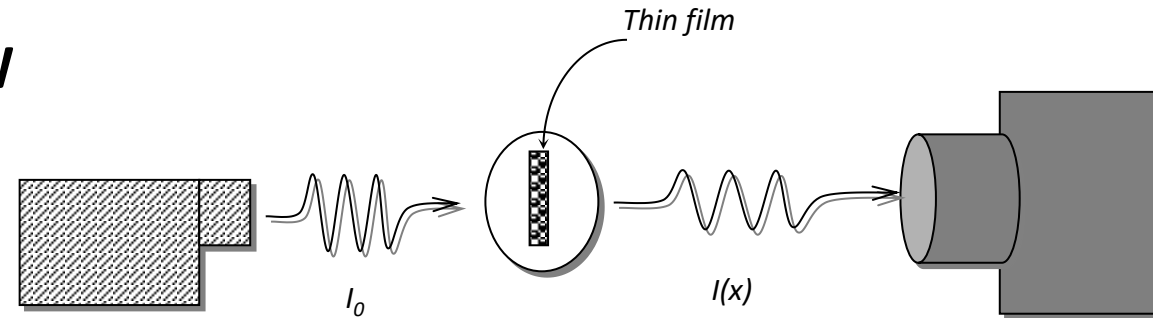
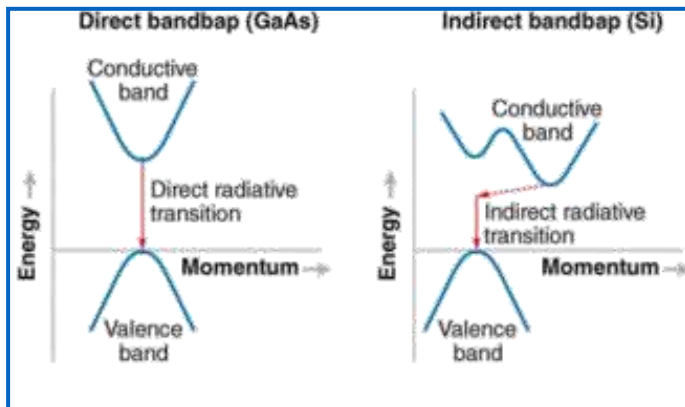
- Ultraviolet radiation stimulates molecular vibrations and electronic transitions.
- Absorption spectroscopy from 160 nm to 780 nm.
- Measurement absorption or transmittance.
- Identification of inorganic and organic species

UV-Visible Spectroscopy

THE BEER-LAMBERT LAW

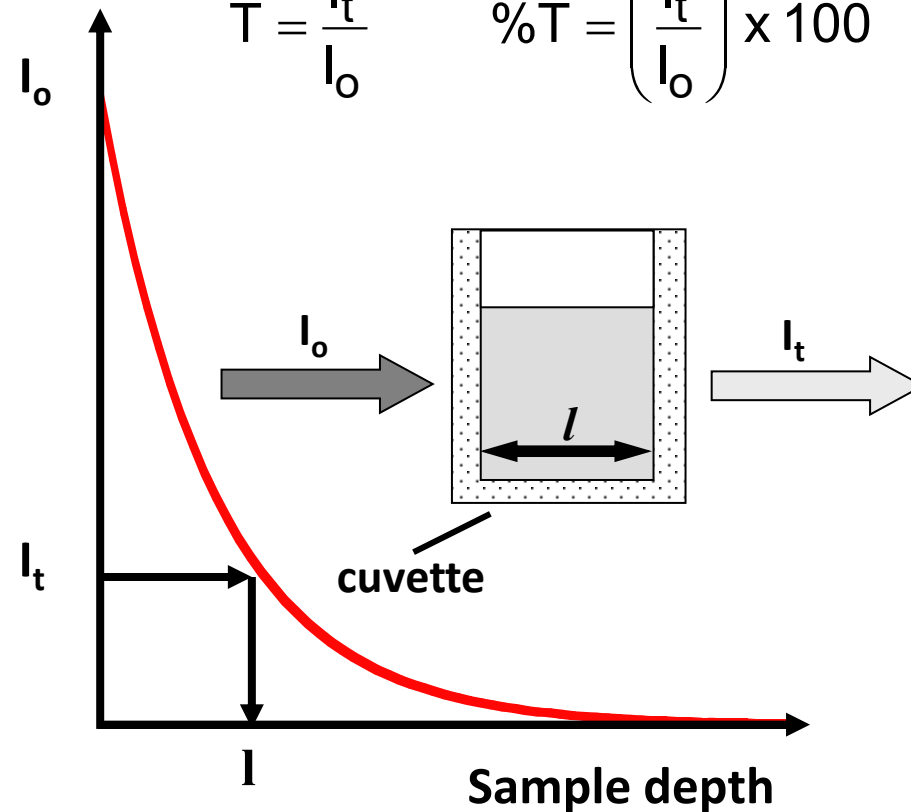
For a light absorbing medium, the light intensity falls exponentially with sample depth.

For a light absorbing medium, the light intensity falls exponentially with increasing sample concentration.

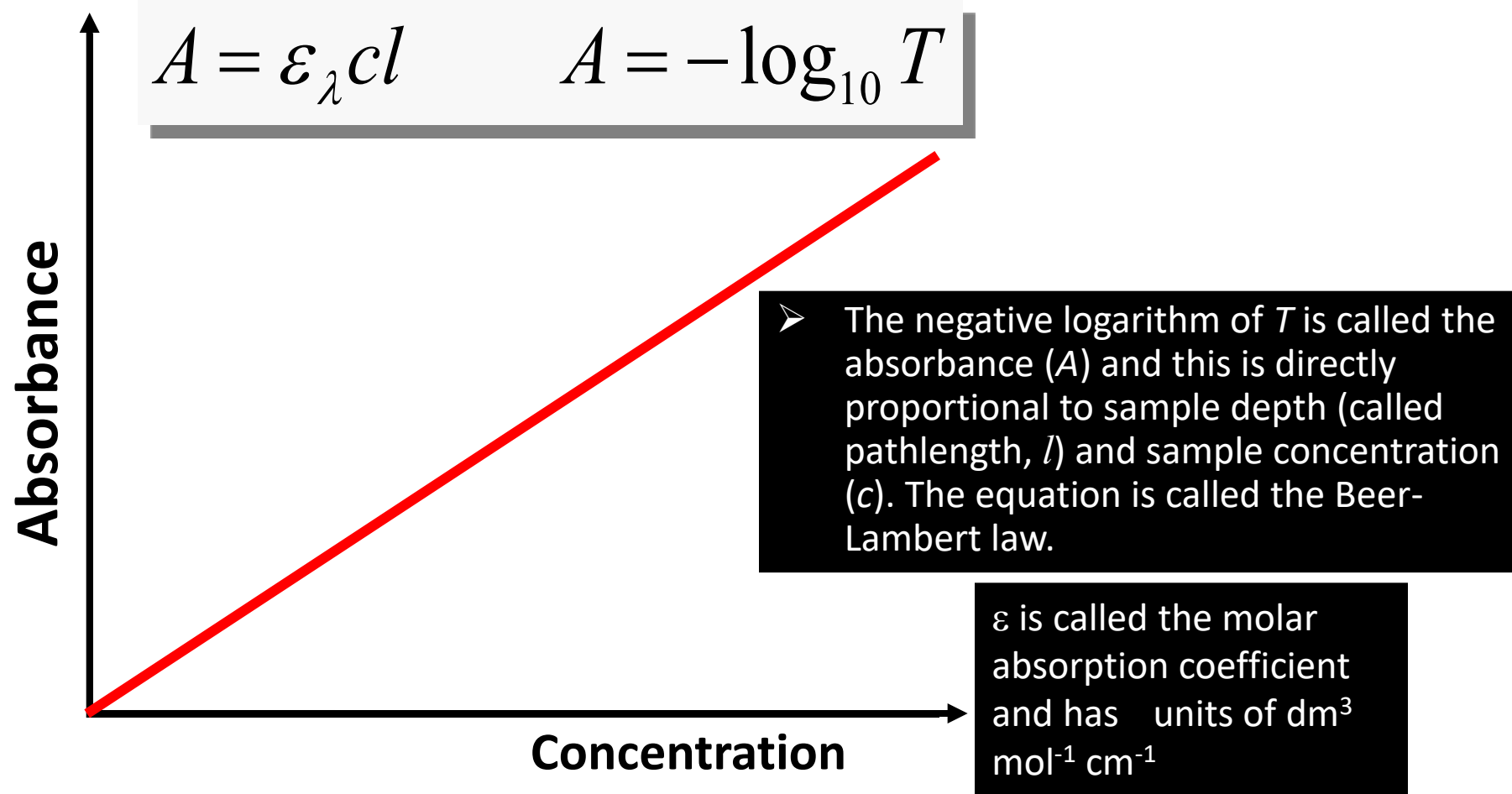


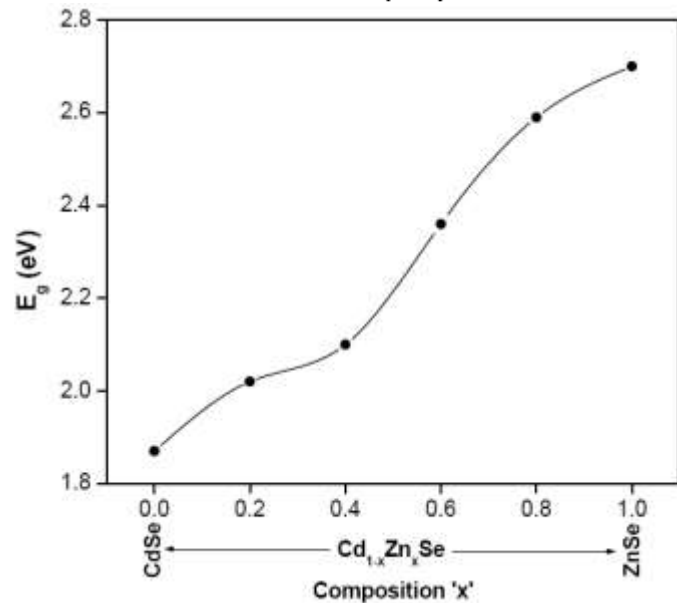
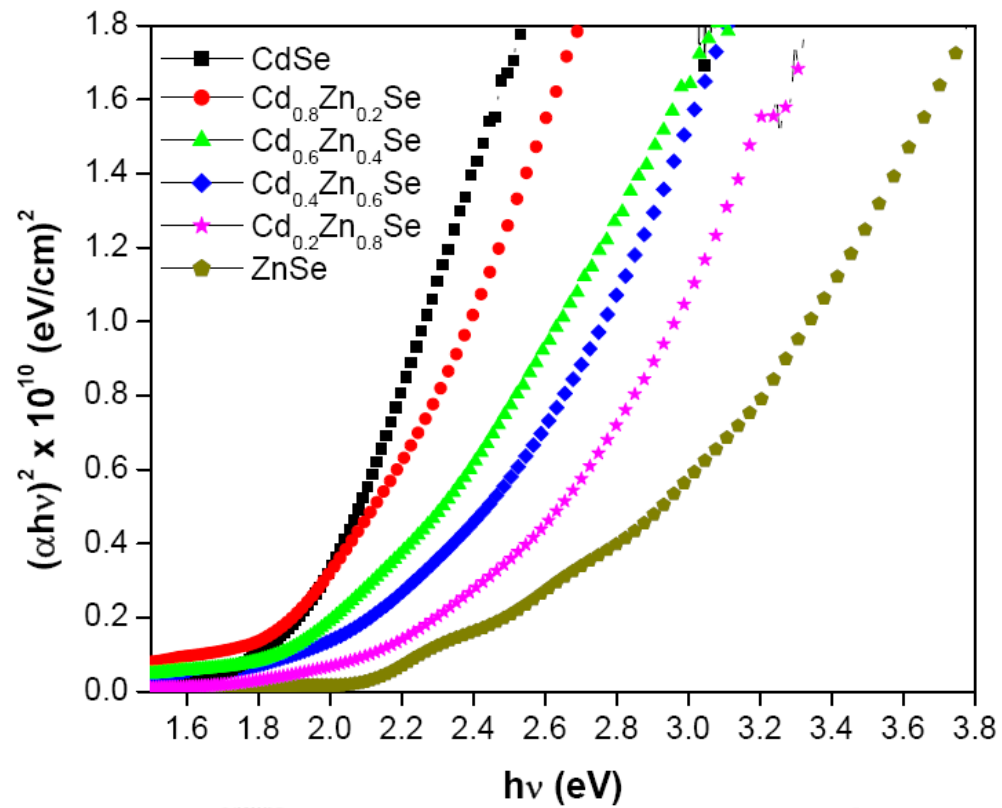
$$T = \frac{I_t}{I_0} \quad \%T = \left(\frac{I_t}{I_0} \right) \times 100$$

light intensity (I)



UV-Visible Spectroscopy



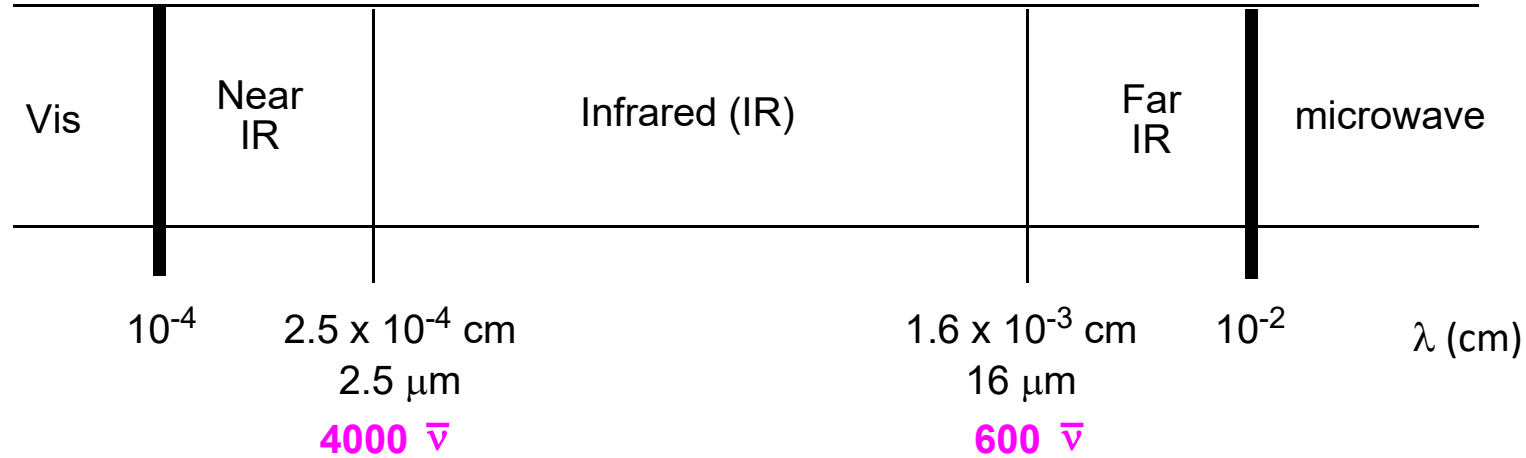


$$\alpha h\nu = A(h\nu - E_g)^n$$

Optical Analysis:

No.	Composition	E_g (eV)
1	CdSe	1.87
2	$\text{Cd}_{0.8}\text{Zn}_{0.2}\text{Se}$	2.02
3	$\text{Cd}_{0.6}\text{Zn}_{0.4}\text{Se}$	2.10
4	$\text{Cd}_{0.4}\text{Zn}_{0.6}\text{Se}$	2.36
5	$\text{Cd}_{0.2}\text{Zn}_{0.8}\text{Se}$	2.59
6	ZnSe	2.70

Introduction to Infrared Spectroscopy

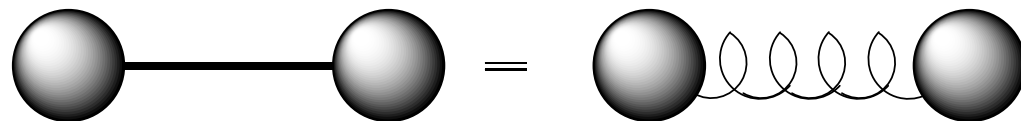


$$E \propto \frac{1}{\lambda}$$

λ is expressed as $\bar{\nu}$ (wavenumber), reciprocal cm (cm^{-1})

$$\bar{\nu} = \frac{1}{\lambda} \quad \text{therefore} \quad E \propto \bar{\nu}$$

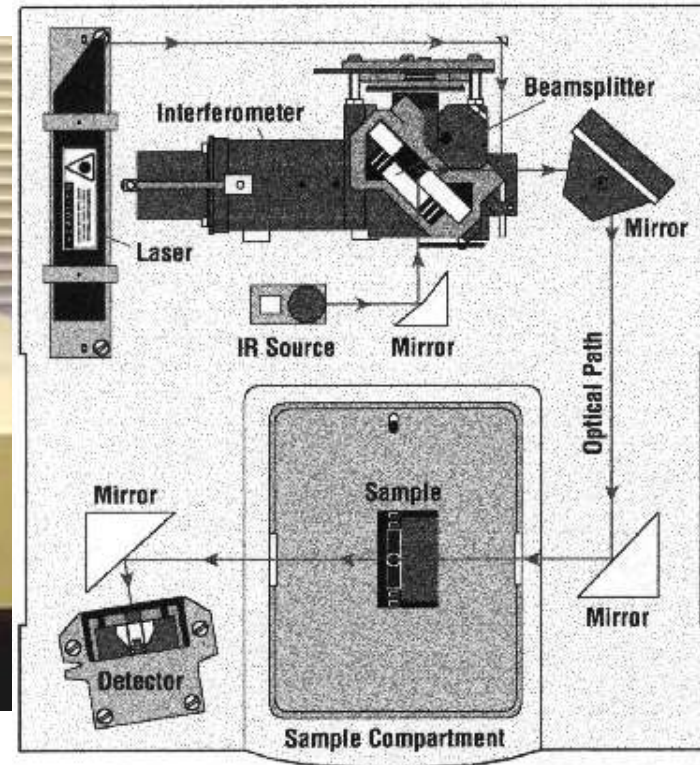
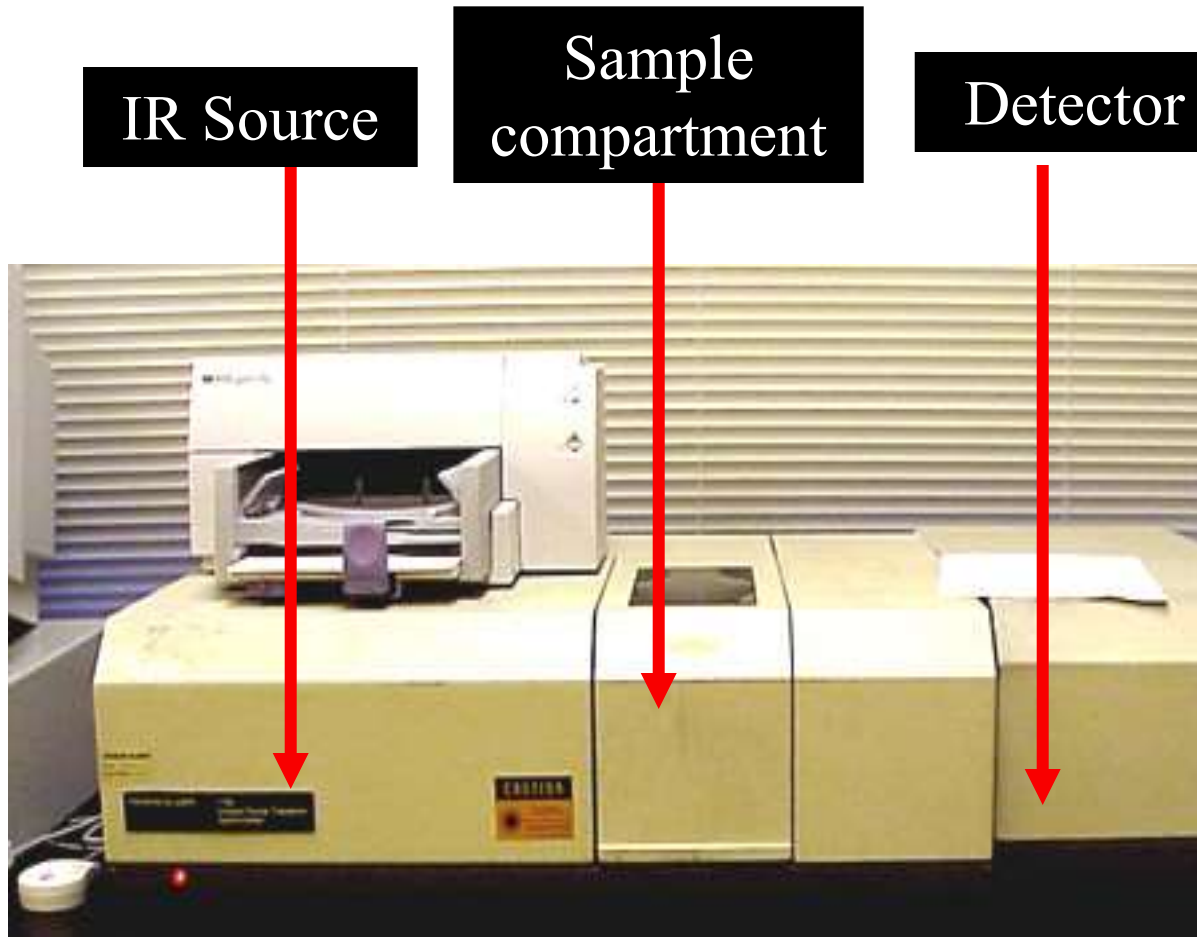
IR radiation causes changes in a molecular vibrations



INFRARED SPECTROSCOPY

- Infrared radiation stimulates molecular vibrations.
- Infrared spectra are traditionally displayed as %T (percent transmittance) versus wave number ($4000\text{-}400\text{ cm}^{-1}$).
- Useful in identifying presence or absence of functional groups.

Infrared Instrumentation

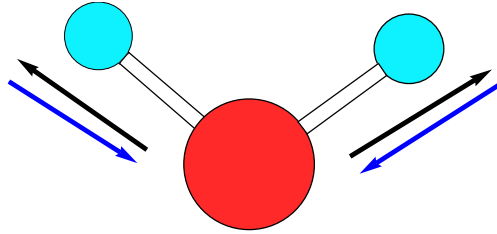


Infrared Instrumentation

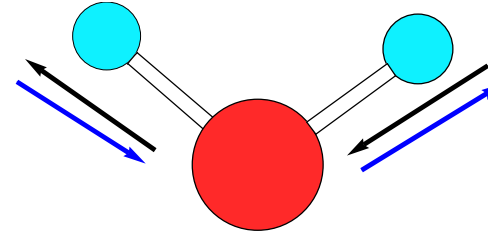
- All modern instruments are Fourier Transform instruments.
- In all transmission experiments radiation from a source is directed through the sample to a detector.
- The measurement of the type and amount of light transmitted by the sample gives information about the structure of the molecules comprising the sample.

Stretch: change in bond length

<http://www2.chem.ucalgary.ca/Flash/photon.html>

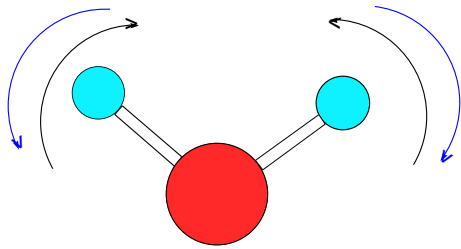


Symmetric stretch

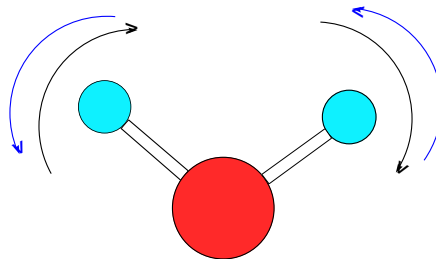


Antisymmetric stretch

Bend: change in bond angle

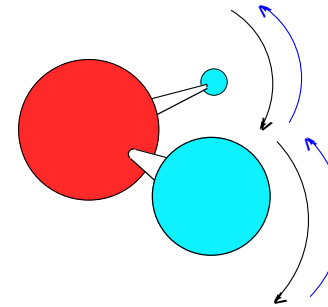


scissoring

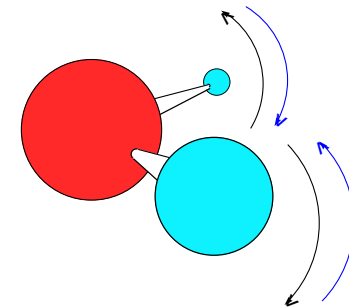


rocking

in-plane bend



wagging



twisting

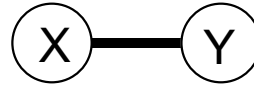
out-of-plane bend

Animation of bond stretches and bends:

<http://wetche.cmbi.ru.nl//organic/vibr/methamjm.html>

Bond Stretch:

Hooke's Law



$$\bar{\nu} = \frac{1}{2 \pi c} \left[\frac{f}{\left(\frac{m_x m_y}{m_x + m_y} \right)} \right]^{\frac{1}{2}}$$

$\bar{\nu}$ = vibrational frequency

c = speed of light

m_x = mass of X

m_y = mass of Y

$\left(\frac{m_x m_y}{m_x + m_y} \right)$ = reduced mass (μ)

f = spring constant; type of
bond between X and Y
(single, double or triple)

$$E \propto \nu \propto \bar{\nu} \propto f$$

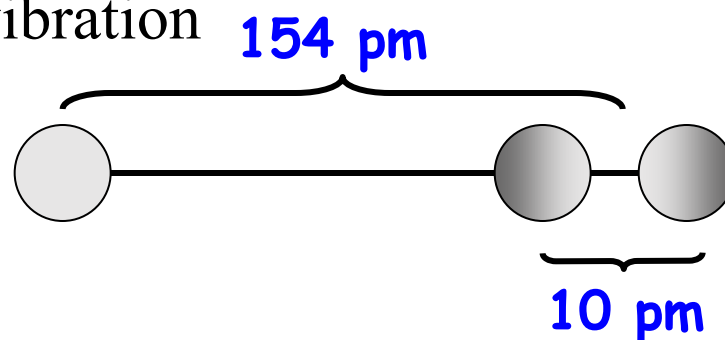
Hooke's law simulation:

<http://www2.chem.ucalgary.ca/Flash/hooke.html>

INFRARED SPECTROSCOPY

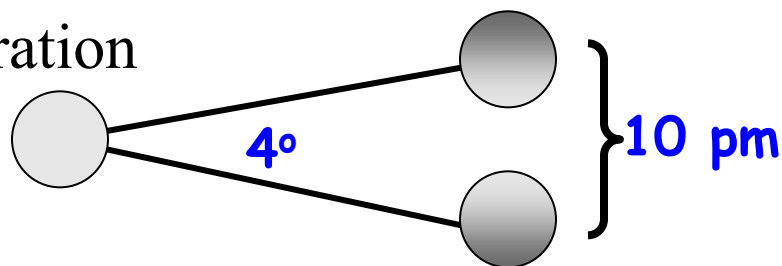
- How much movement occurs in the vibration of a C-C bond?

stretching vibration



For a C-C bond with a bond length of 154 pm, the variation is about 10 pm.

bending vibration



For C-C-C bond angle a change of 4° is typical. This moves a carbon atom about 10 pm.

INFRARED SPECTROSCOPY

POSITION	REDUCED MASS BOND STRENGTH (STIFFNESS)	LIGHT ATOMS HIGH FREQUENCY STRONG BONDS HIGH FREQUENCY
STRENGTH	CHANGE IN 'POLARITY'	STRONGLY POLAR BONDS GIVE INTENSE BANDS
WIDTH	HYDROGEN BONDING	STRONG HYDROGEN BONDING GIVES BROAD BANDS

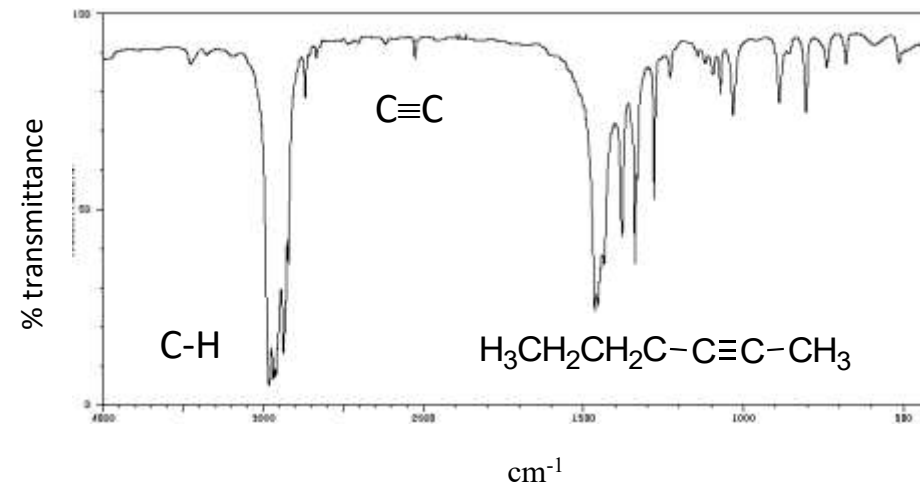
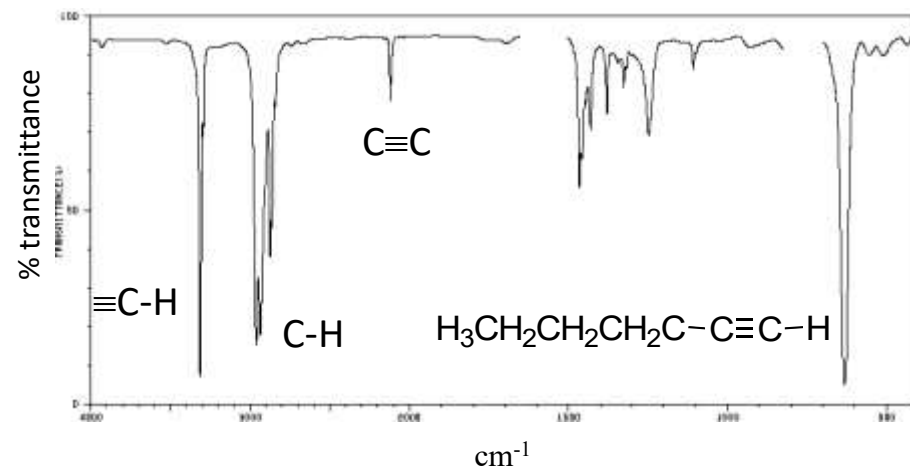
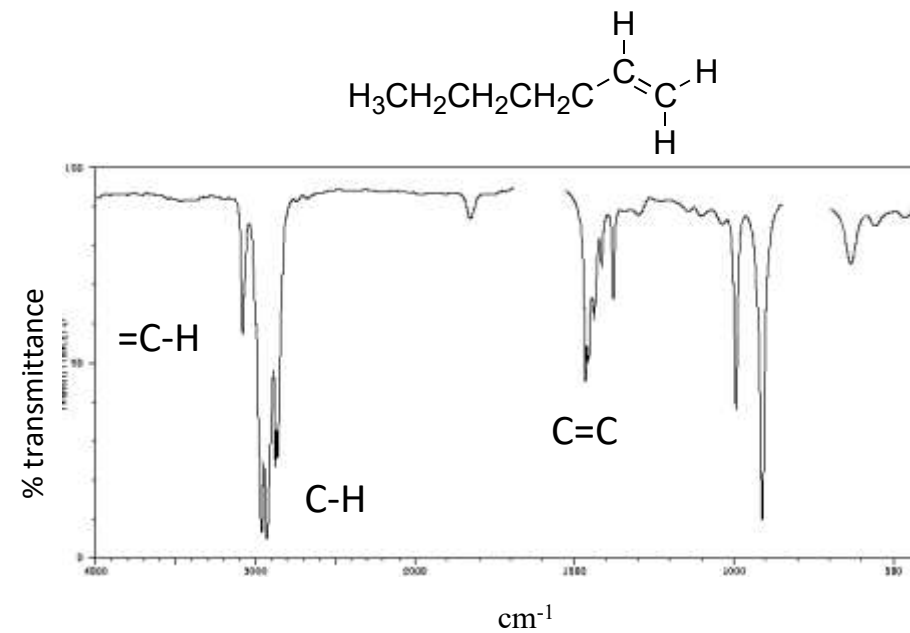
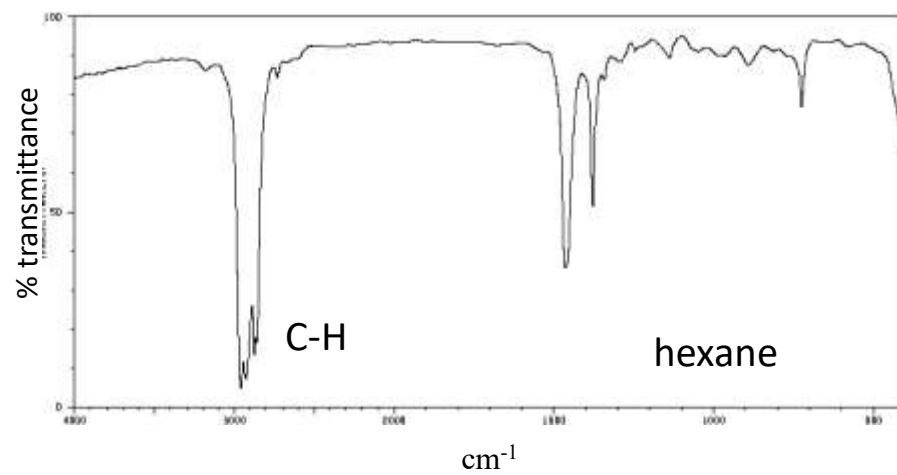
INFRARED SPECTROSCOPY

- Interpretation of an Infrared Spectra:

organic molecules contain many atoms. As a result, there are many stretching and bending modes- IR spectra have many absorption bands

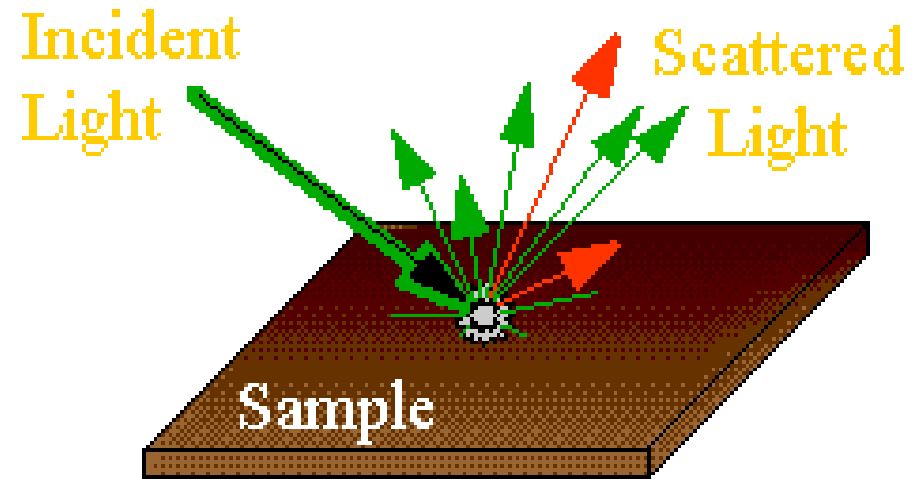
- In general

Bond →	C-H	C-D	C-O	C-Cl
ν/cm^{-1} →	3000	2200	1100	700
Bond →	C≡O	C=O	C-O	
ν/cm^{-1} →	2143	1715	1100	



Raman Spectroscopy

Laser 4880 Å



When radiation passes through a transparent medium, the species present scatter a fraction of the beam in all directions.

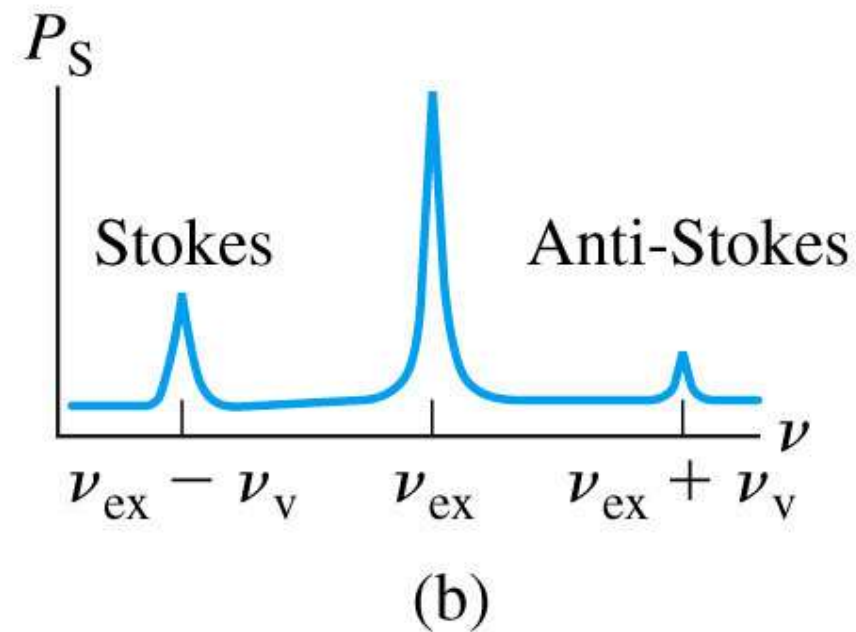
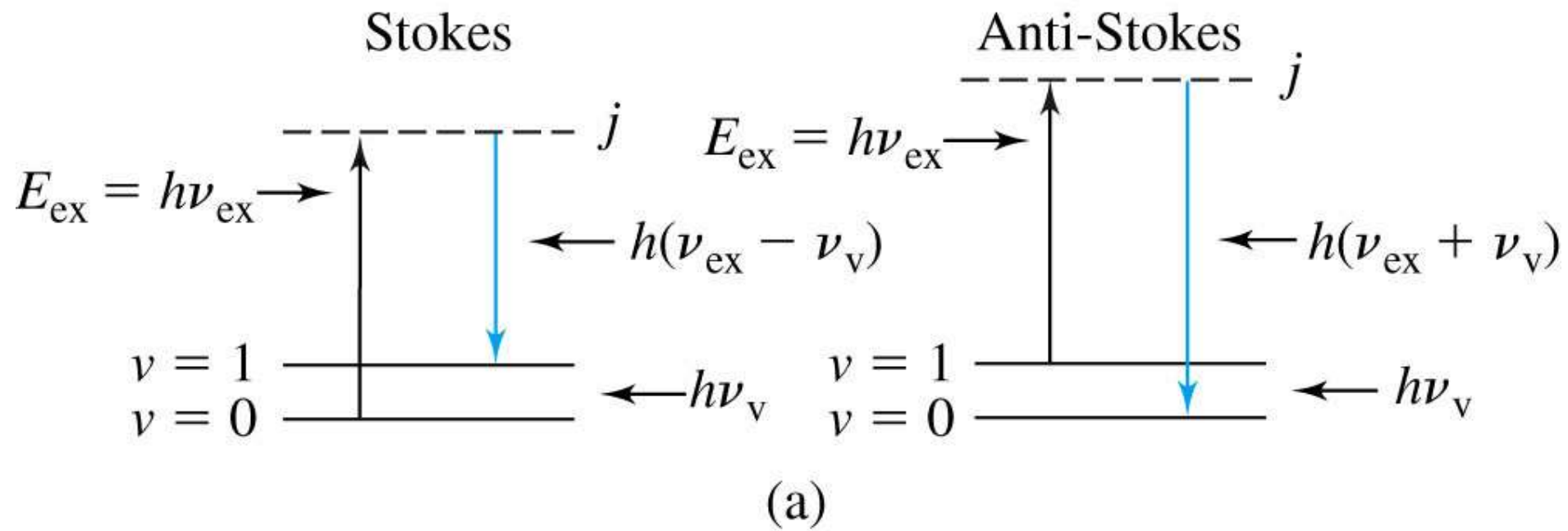
In 1928, the Indian physicist C. V. Raman discovered that the visible wavelength of a small fraction of the radiation scattered by certain molecules differs from that of the incident beam and furthermore that the shifts in wavelength depend upon the chemical structure of the molecules responsible for the scattering.

Excitation of Raman Spectra

A Raman spectrum can be obtained by irradiating a sample of carbon tetrachloride with an intense beam of an argon ion laser having a wavelength of 488.0 nm (20492 cm^{-1}).

The emitted radiation is of three types:

1. **Stokes** scattering
2. **Anti-stokes** scattering
3. **Rayleigh** scattering



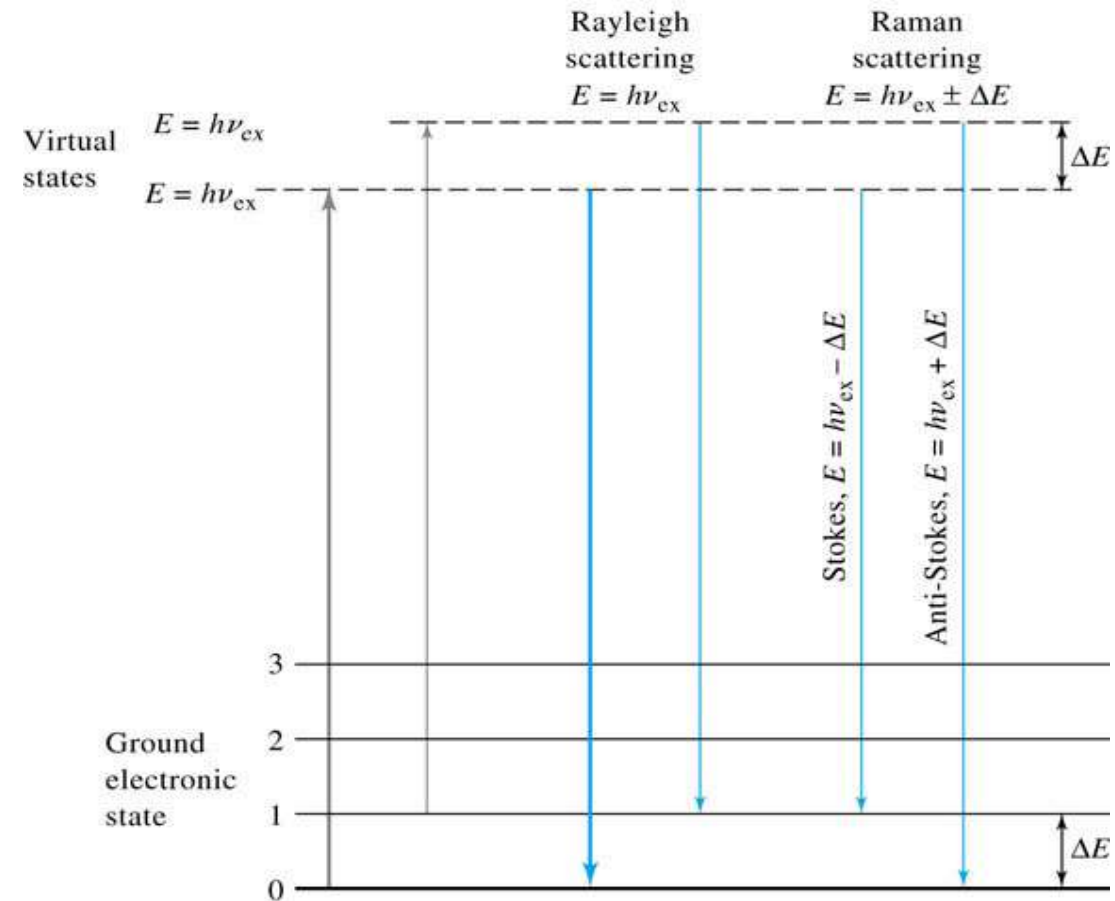
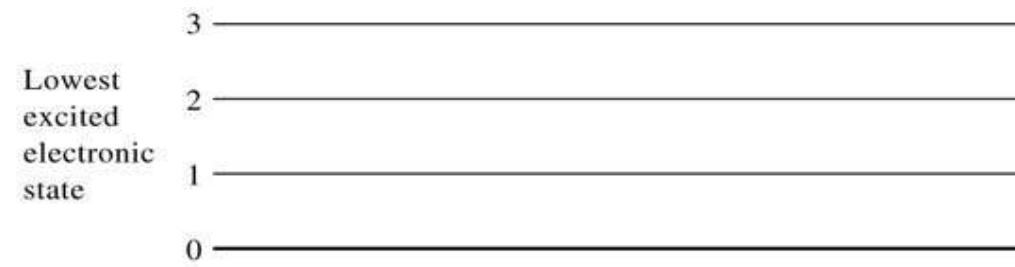
Excitation of Raman Spectra

The abscissa of Raman spectrum is the wavenumber shift $\Delta\nu$, which is defined as the difference in wavenumbers (cm^{-1}) between the observed radiation and that of the source.

For CCl_4 three peaks are found on both sides of the Rayleigh peak and that the pattern of shifts on each side is identical.

Anti-Stokes lines are appreciably less intense than the corresponding Stokes lines. For this reason, only the Stokes part of a spectrum is generally used.

The magnitude of Raman shifts are independent of the wavelength of excitation.



Mechanism of Raman and Rayleigh Scattering

The middle set of arrows depicts the changes that produce Rayleigh scattering.

The energy changes that produce stokes and anti-Stokes emission are depicted on the right.

The two differ from the Rayleigh radiation by frequencies corresponding to $\pm\Delta E$, the energy of the first vibrational level of the ground state.

If the bond were infrared active, the energy of its absorption would also be ΔE . Thus, the Raman frequency shift and the infrared absorption peak frequency are identical.

Mechanism of Raman and Rayleigh Scattering

The relative populations of the two upper energy states are such that Stokes emission is much favored over anti-Stokes.

Rayleigh scattering has a considerably higher probability of occurring than Raman because the most probable event is the energy transfer to molecules in the ground state and reemission by the return of these molecules to the ground state.

The ratio of anti-Stokes to Stokes intensities will increase with temperature because a larger fraction of the molecules will be in the first vibrationally excited state under these circumstances.

Raman vs. I.R

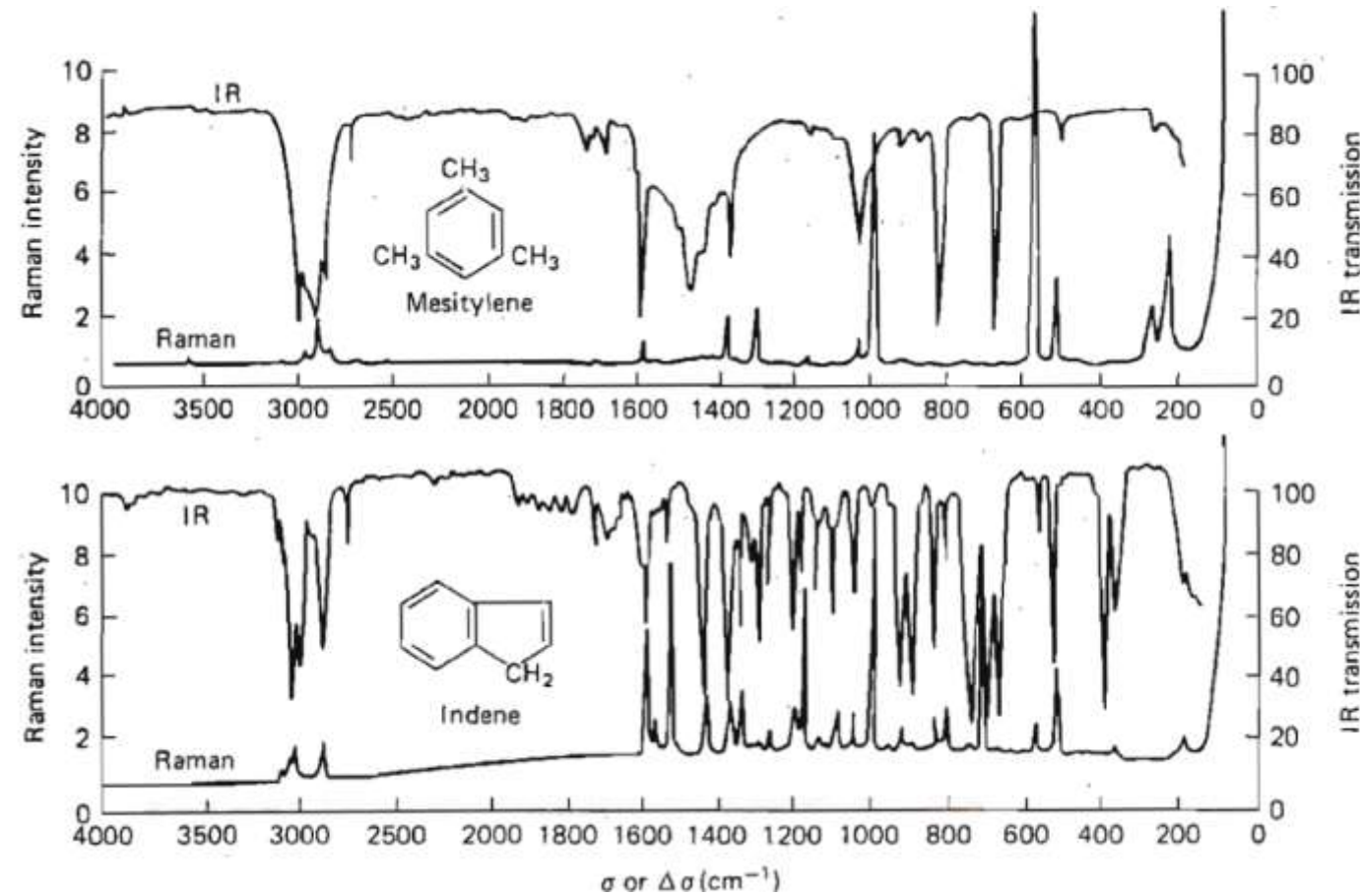
For a given bond, the energy shifts observed in a Raman experiment should be identical to the energies of its infrared absorption bands, provided that the vibrational modes involved are active toward both infrared absorption and Raman scattering.

The differences between a Raman spectrum and an infrared spectrum are not surprising.

Infrared absorption requires that a vibrational mode of the molecule have a change in dipole moment or charge distribution associated with it.

Raman Spectroscopy: General

- IR and Raman are both useful for Fingerprinting



- Symmetry dictates which are active in Raman and IR

Raman Spectroscopy: Summary

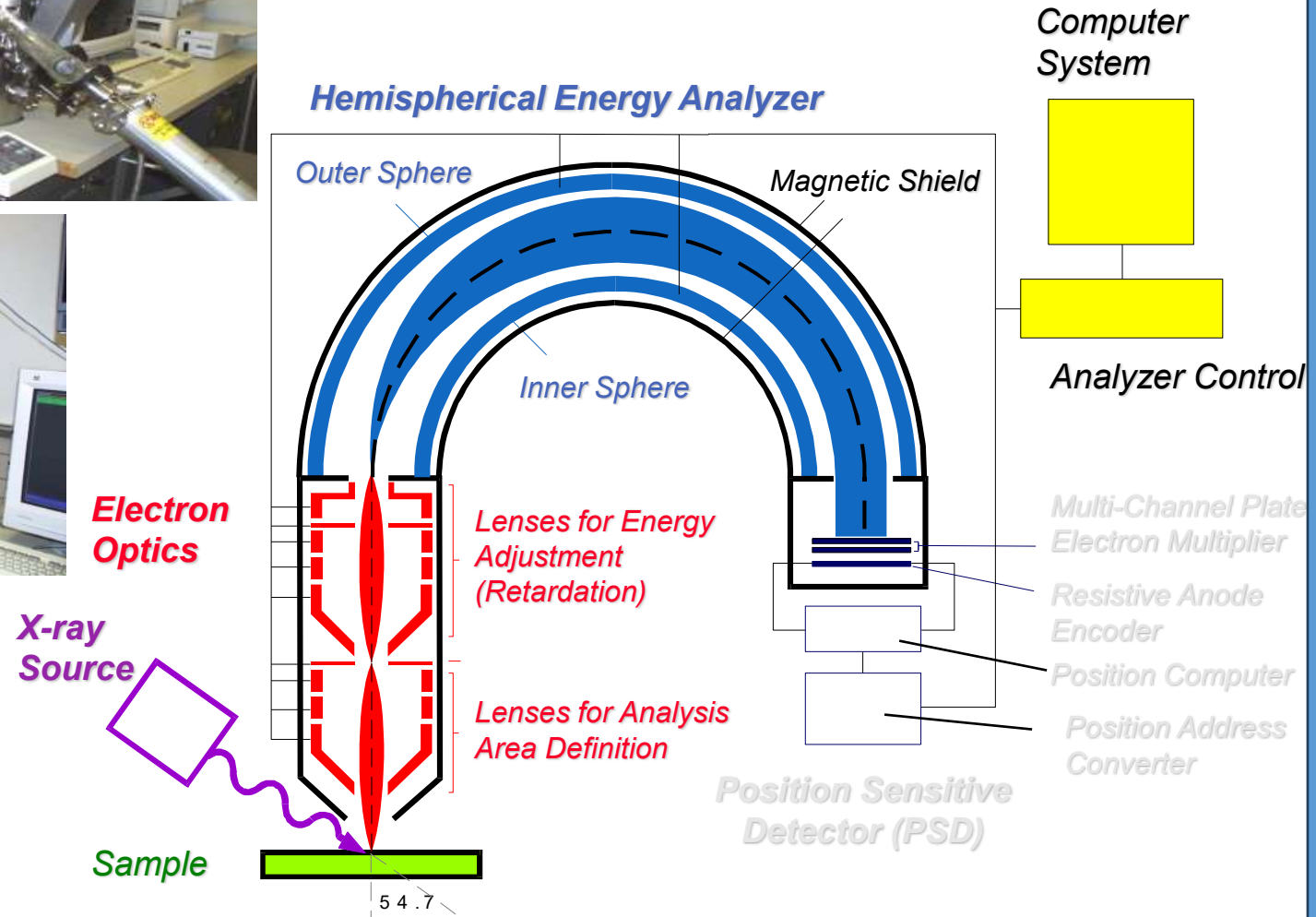
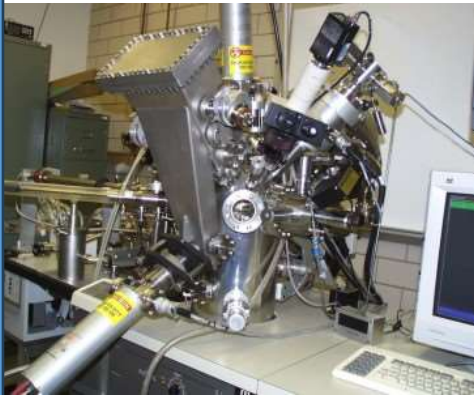
1. Raman is a vibrational spectroscopy akin to IR
 - Good for fingerprinting, probing molecular symmetry
2. Scattering-based, not transmission/reflection
 - Means no need for fancy sample preparation...gas, liquid, or solid
 - Virtually always use anti-Stokes lines due to stronger signal
3. You need to pick excitation energy (laser line)
 - 785 nm: Fluorescence less probable; Lower Raman signal
 - 514 nm: Fluorescence more probable; Resonance more likely; Higher signal
4. Other things not talked about
 - SERS: Surface Enhanced Raman Spectroscopy
 - Quantum origins of selection rules and scattering cross-section

X-ray Photoemission Spectroscopy (XPS) and Auger Electron Spectroscopy (AES) technique

Introduction

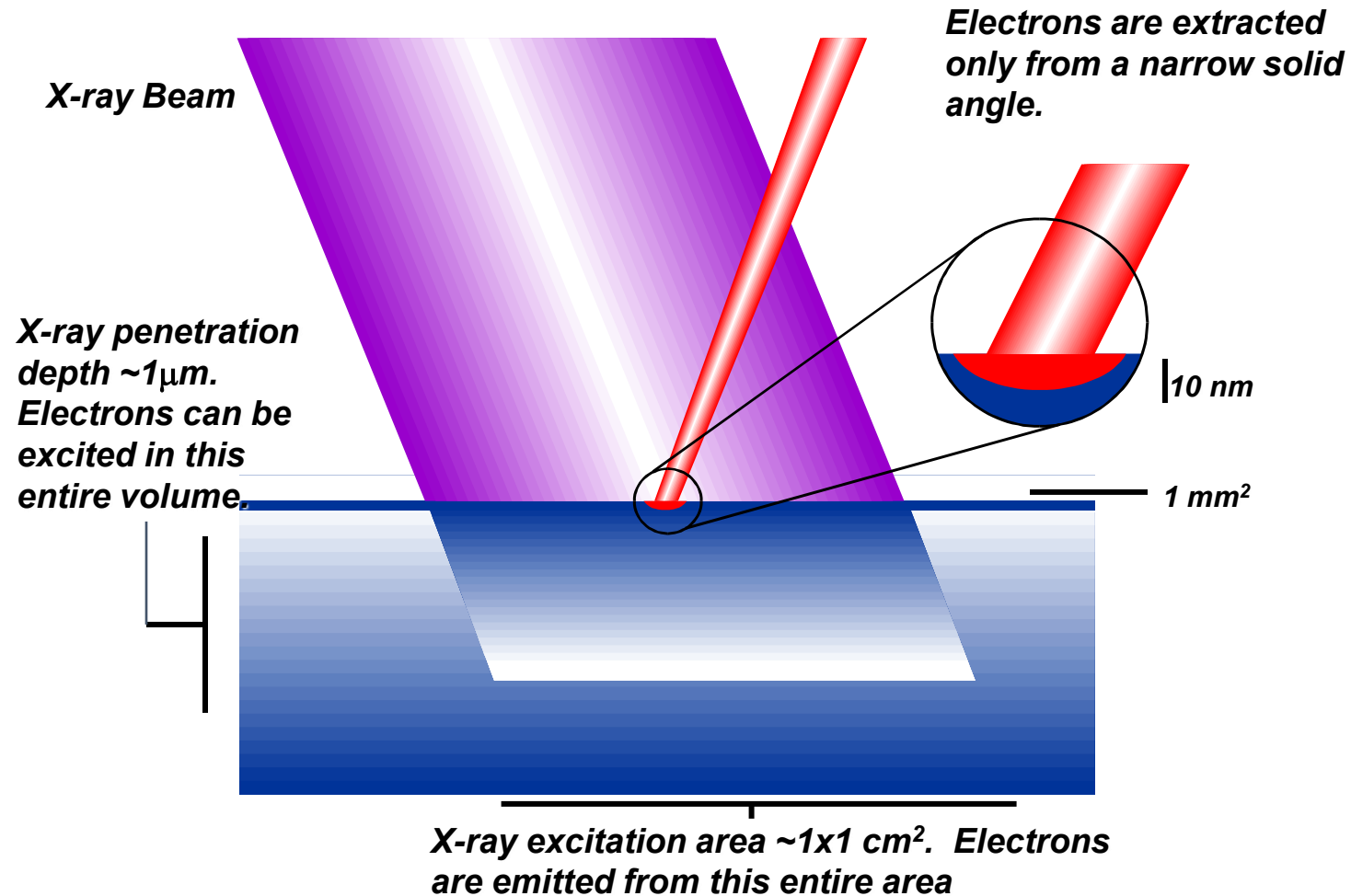
- Electron spectroscopy (ES) is a technique that uses characteristic electrons emitted from solid for elemental analysis, not for imaging as in electron microscopy.
- The characteristic electrons (either Auger electrons or photoelectrons) exhibit characteristics energy levels, revealing the nature of chemical elements in specimen being examined.
- Auger or photoelectrons can only escape from the uppermost atomic layers of solid (a depth of 10 nm or less) because their energies are relatively low (generally 20-2000 eV).
- Thus, ES is a technique for surface analysis.
- There two types of ES: Auger electron spectroscopy (AES) and X-ray photoelectron spectroscopy (XPS).
- Auger electrons and photoelectrons are different in their physical origins, but both types of electrons carry similar information about chemical elements in material surfaces.

X-ray Photoelectron Spectrometer



X-ray Photoelectron Spectroscopy

Small Area Detection

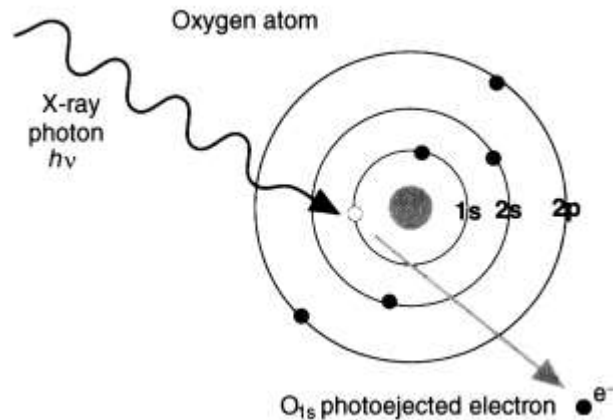


XPS technique is based on Einstein's idea about the photoelectric effect, developed around 1905.

-The concept of photons was used to describe the ejection of electrons from a surface when photons were impinged upon it

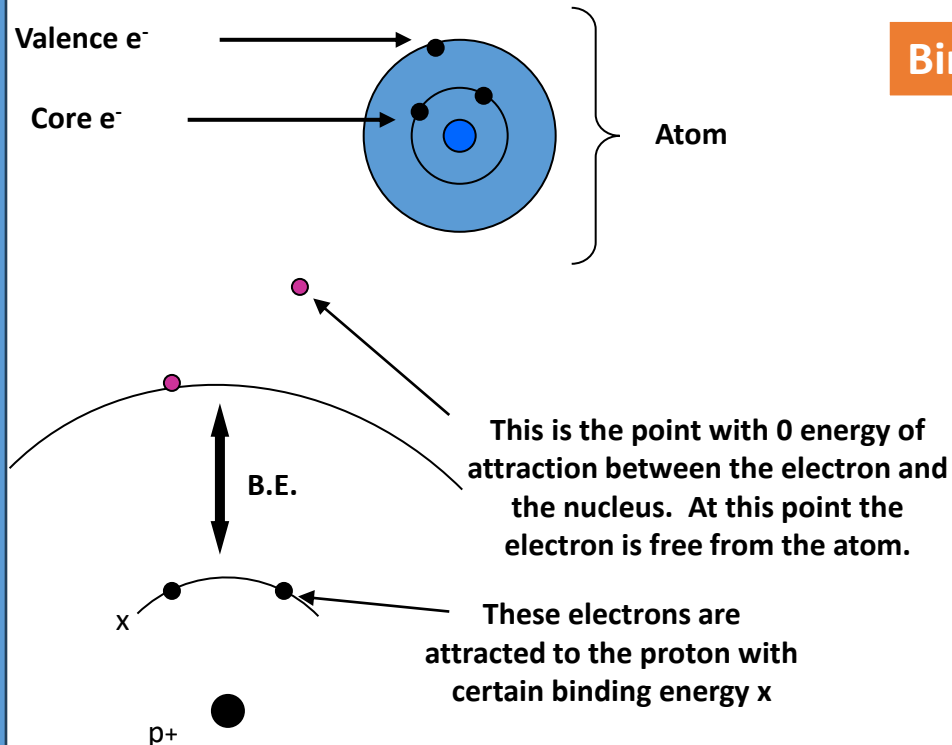
X-Rays

- Irradiate the sample surface, hitting the core electrons (e^-) of the atoms.
- The X-Rays penetrate the sample to a depth on the order of a micrometer.
- Useful e^- signal is obtained only from a depth of around 10 to 100 Å on the surface.
- The X-Ray source produces photons with certain energies:
 - MgK_α photon with an energy of 1253.6 eV
 - AlK_α photon with an energy of 1486.6 eV
- Normally, the sample will be radiated with photons of a single energy (MgK_α or AlK_α). This is known as a monoenergetic X-Ray beam.



Why the Core Electrons?

- An electron near the Fermi level is far from the nucleus, moving in different directions all over the place, and will not carry information about any single atom.
 - Fermi level is the highest energy level occupied by an electron in a neutral solid at absolute 0 temperature.
 - Electron binding energy (BE) is calculated with respect to the Fermi level.
- The core e⁻s are local close to the nucleus and have binding energies characteristic of their particular element.
- The core e⁻s have a higher probability of matching the energies of AlK_α and MgK_α.



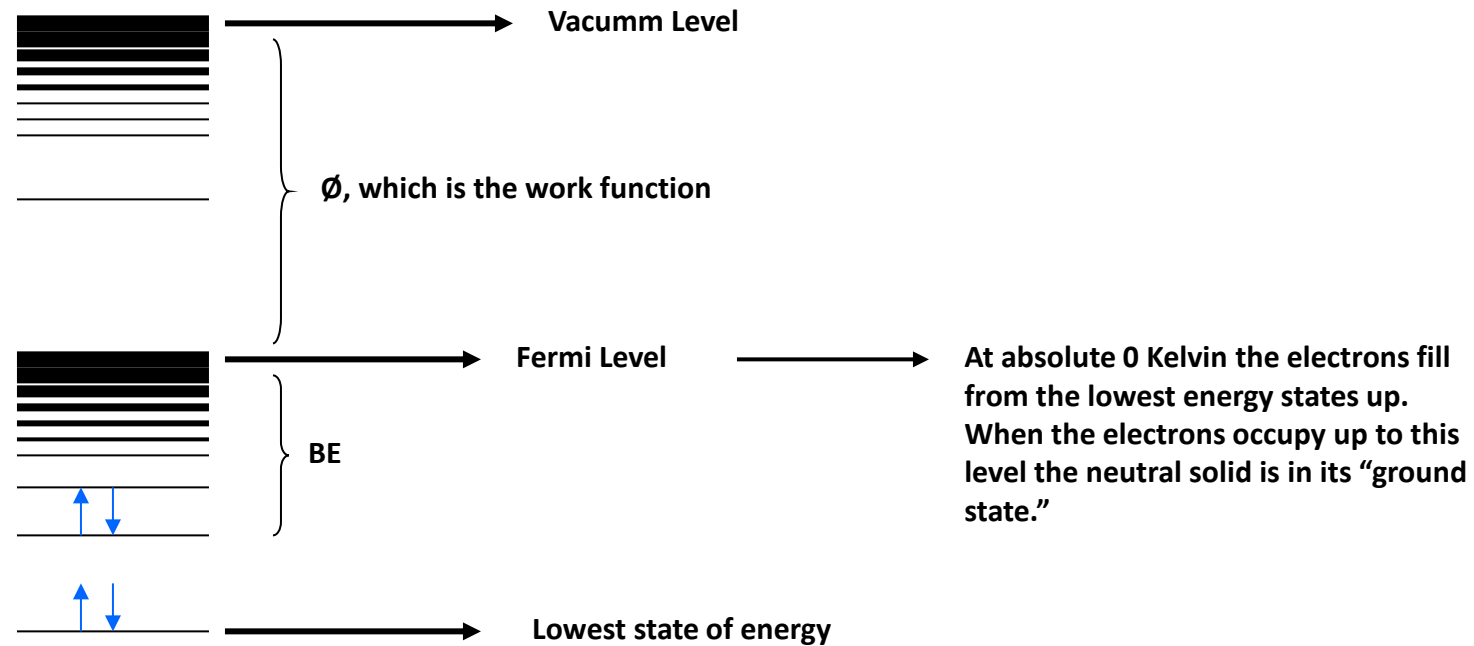
Binding Energy (BE)

BE is characteristic of the core electrons for each element.

The BE is determined by the attraction of the electrons to the nucleus.

If an electron with energy x is pulled away from the nucleus, the attraction between the electron and the nucleus decreases and the BE decreases. Eventually, there will be a point when the electron will be free of the nucleus.

Energy Levels



How Does XPS Technology Work?

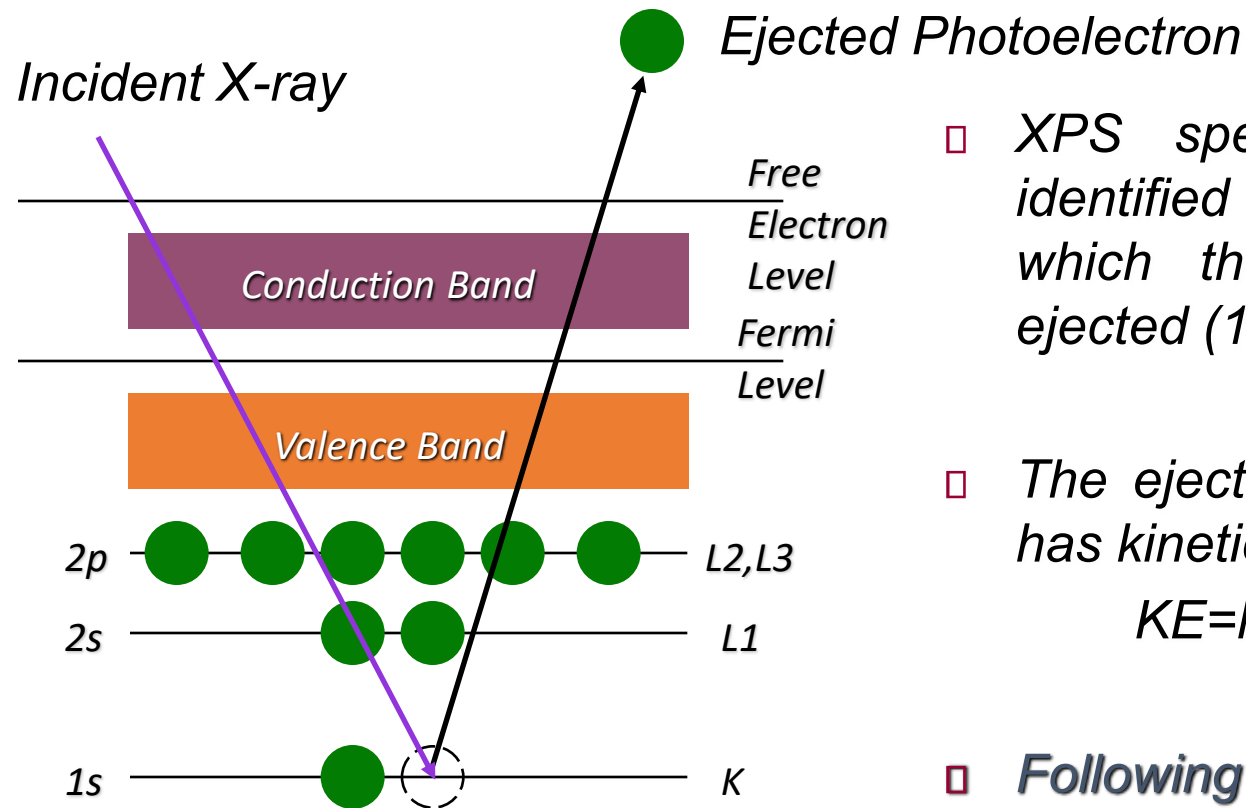
- A monoenergetic x-ray beam emits photoelectrons from the surface of the sample.
- Ultrahigh vacuum (UHV) environment to eliminate excessive surface contamination.
- The X-Rays either of two energies:
 - Al K_{α} (1486.6eV)
 - Mg K_{α} (1253.6 eV)
- Cylindrical Mirror Analyzer (CMA) measures the KE of emitted e⁻s.
- The x-ray photons: The penetration is about a micrometer of the sample
- The spectrum plotted by the computer from the analyzer signal.
- The binding energies can be determined from the peak positions and the elements present in the sample identified.
- The XPS spectrum contains information only about the top 10 - 100 Å of the sample.

Why Does XPS Need UHV?

<i>Degree of Vacuum</i>	<i>Pressure Torr</i>
<i>Low Vacuum</i>	10^2
<i>Medium Vacuum</i>	10^{-1}
<i>High Vacuum</i>	10^{-4}
<i>Ultra-High Vacuum</i>	10^{-8} 10^{-11}

- Contamination of surface
 - **XPS is a surface sensitive technique.**
 - Contaminates will produce an XPS signal and lead to incorrect analysis of the surface of composition.
- The pressure of the vacuum system is $\leq 10^{-9}$ Torr
- Removing contamination
 - To remove the contamination the sample surface is bombarded with argon ions ($\text{Ar}^+ = 3\text{KeV}$).
 - heat and oxygen can be used to remove hydrocarbons
- The XPS technique could cause damage to the surface, but it is negligible.
- ***Increase the mean free path for electrons, ions and photons.***

The Photoelectric Process



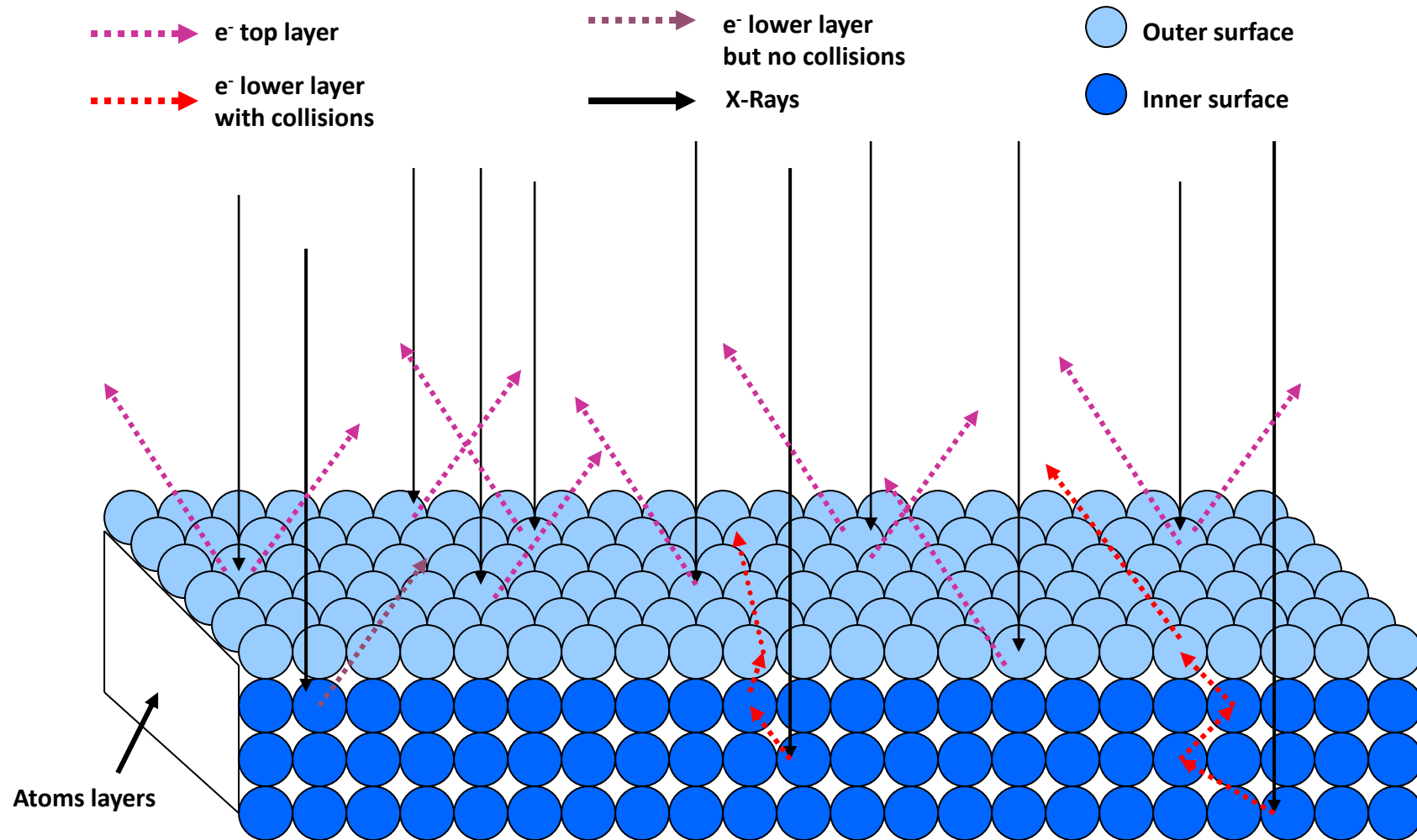
□ XPS spectral lines are identified by the shell from which the electron was ejected (1s, 2s, 2p, etc.).

□ The ejected photoelectron has kinetic energy:

$$KE = h\nu - BE - \Phi$$

□ Following this process, the atom will release energy by the emission of an Auger Electron.

X-Rays on the Surface



X-Rays on the Surface

The X-Rays will penetrate to the core e^- of the atoms in the sample.

Some e^- s are going to be released without any problem giving the Kinetic Energies (KE) characteristic of their elements.

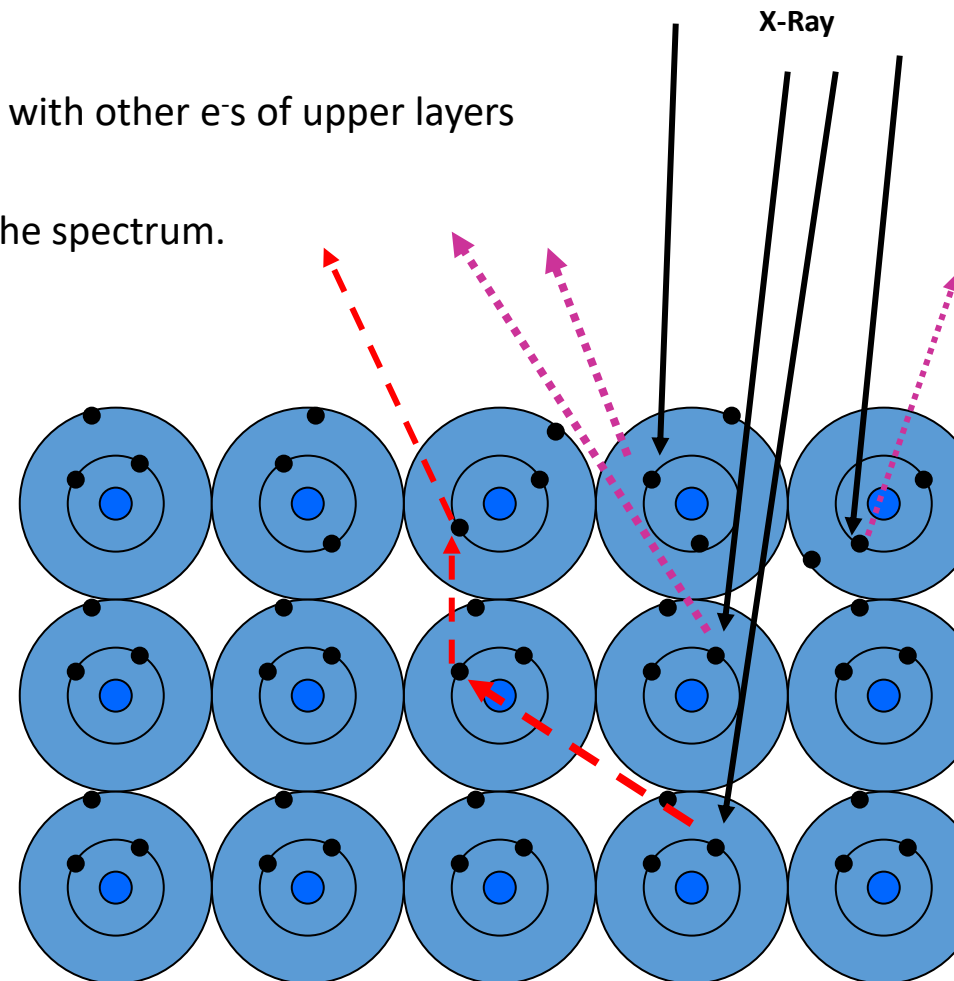
Other e^- s will come from inner layers and collide with other e^- s of upper layers

- These e^- will be lower in lower energy.
- They will contribute to the noise signal of the spectrum.

.....→ Electron without collision

---→ Electron with collision

The noise signal comes from the electrons that collide with other electrons of different layers. The collisions cause a decrease in energy of the electron and it no longer will contribute to the characteristic energy of the element.



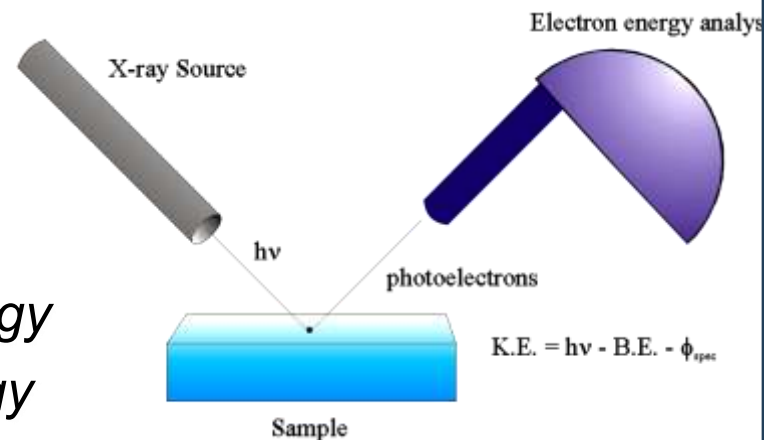
XPS Energy Scale- Binding energy

$$BE = h\nu - KE - \Phi_{spec}$$

Where: **BE** = Electron Binding Energy

KE = Electron Kinetic Energy

Φ_{spec} = Spectrometer Work Function

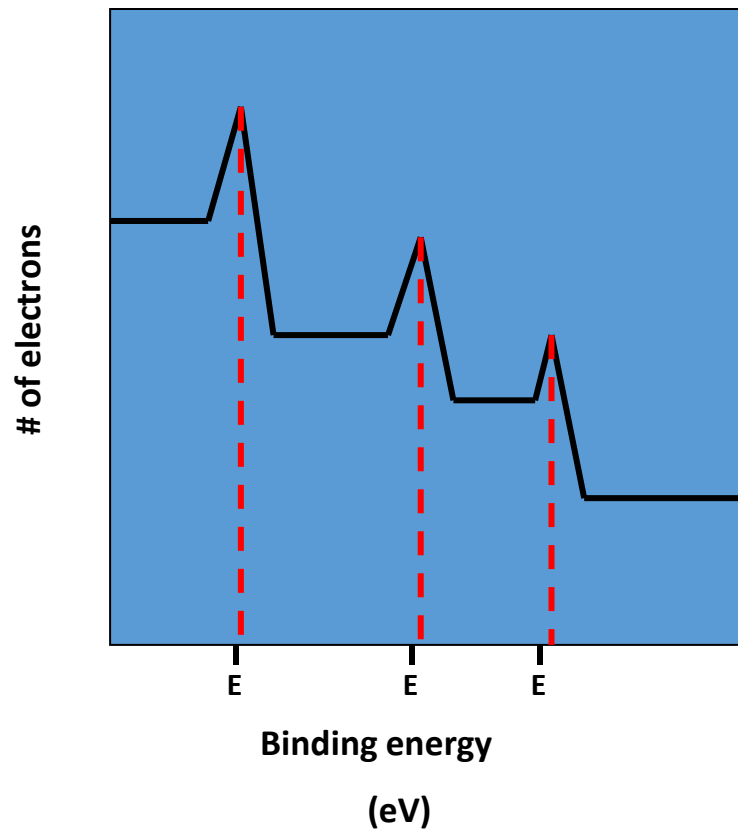


Photoelectron line energies: **Not Dependent** on photon energy.

Auger electron line energies: **Dependent** on photon energy.

The binding energy scale was derived to make uniform comparisons of chemical states straight forward.

KE versus BE



KE can be plotted depending on BE

Each peak represents the amount of e^- s at a certain energy that is characteristic of some element.

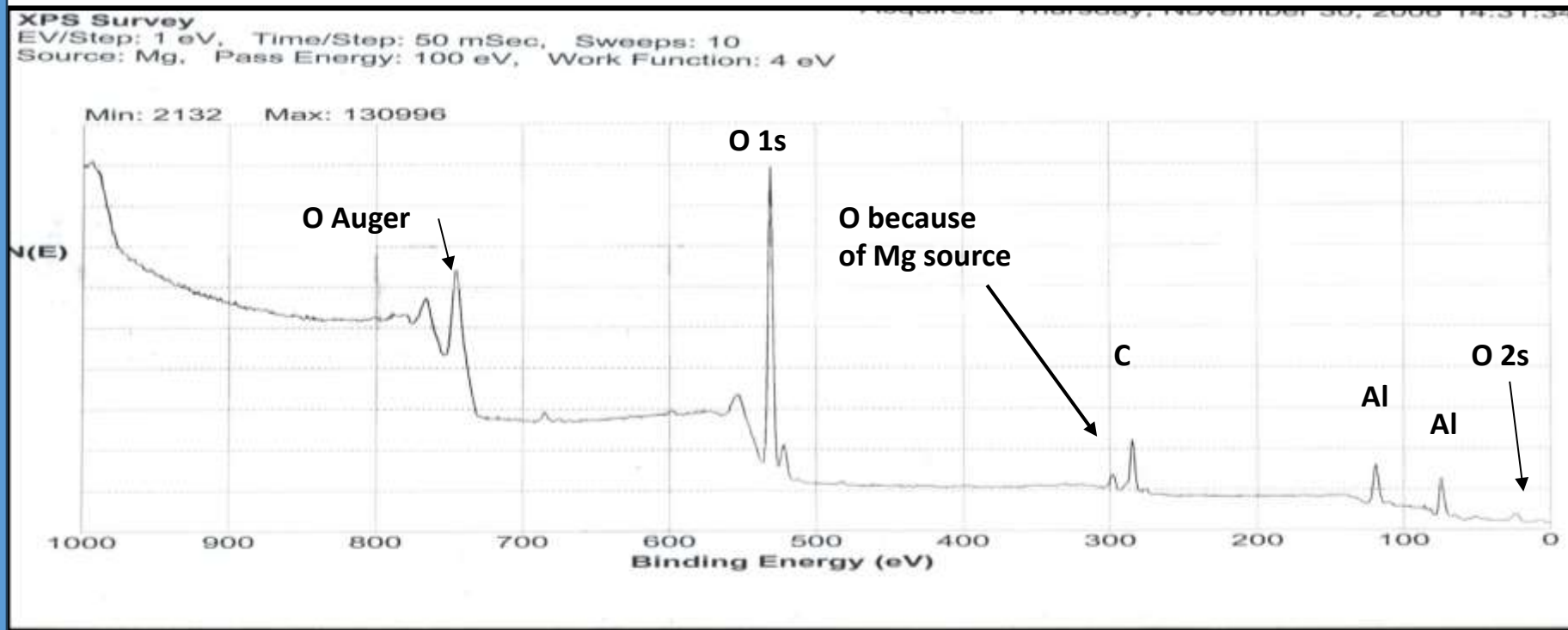
BE increase from right to left



KE increase from left to right

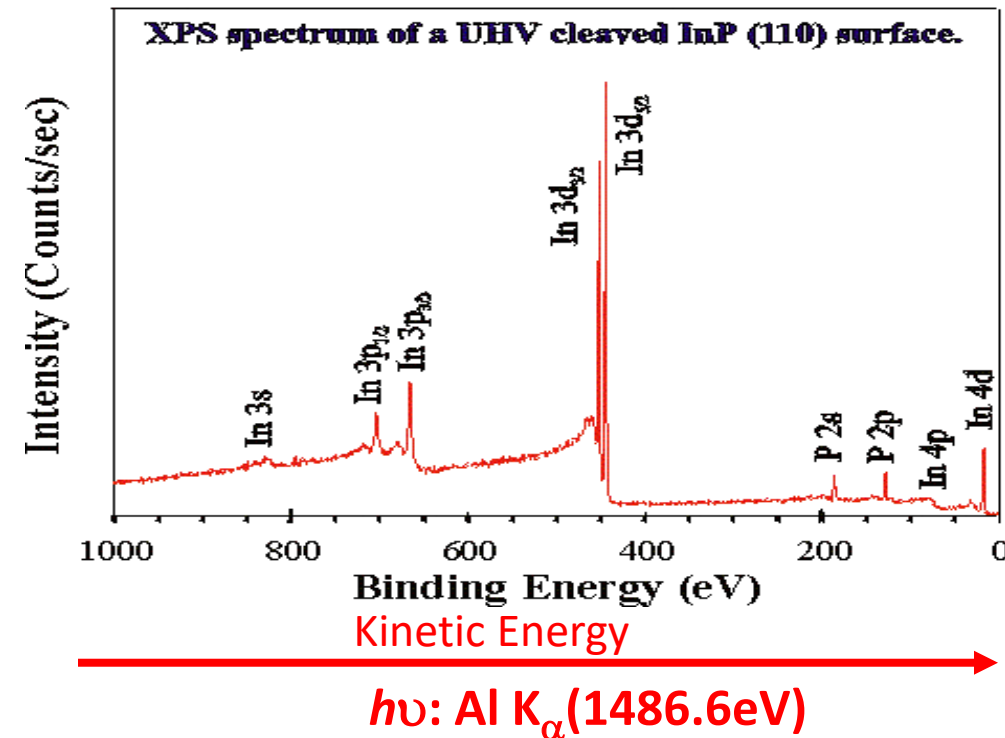
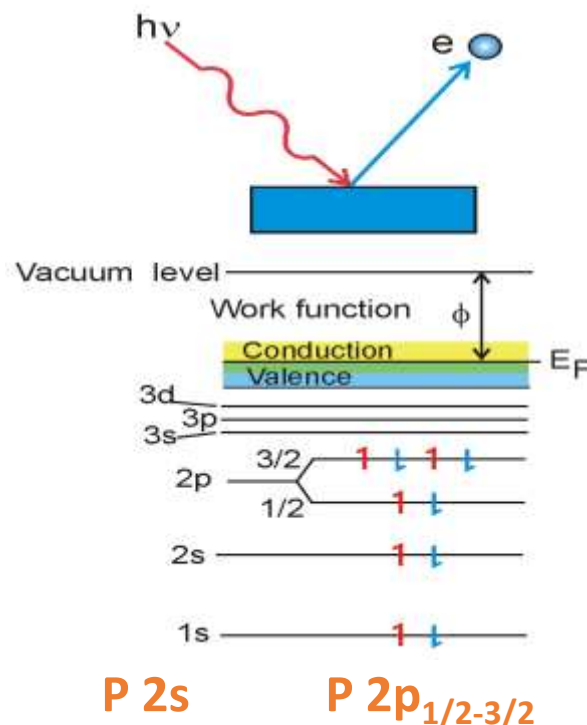
XPS Spectrum

- The XPS peaks are sharp.
- In a XPS graph it is possible to see Auger electron peaks.
- The Auger peaks are usually wider peaks in a XPS spectrum.



Identification of XPS Peaks

- The plot has characteristic peaks for each element found in the surface of the sample.
- There are tables with the KE and BE already assigned to each element.
- After the spectrum is plotted you can look for the designated value of the peak energy from the graph and find the element present on the surface.



Quantitative Analysis

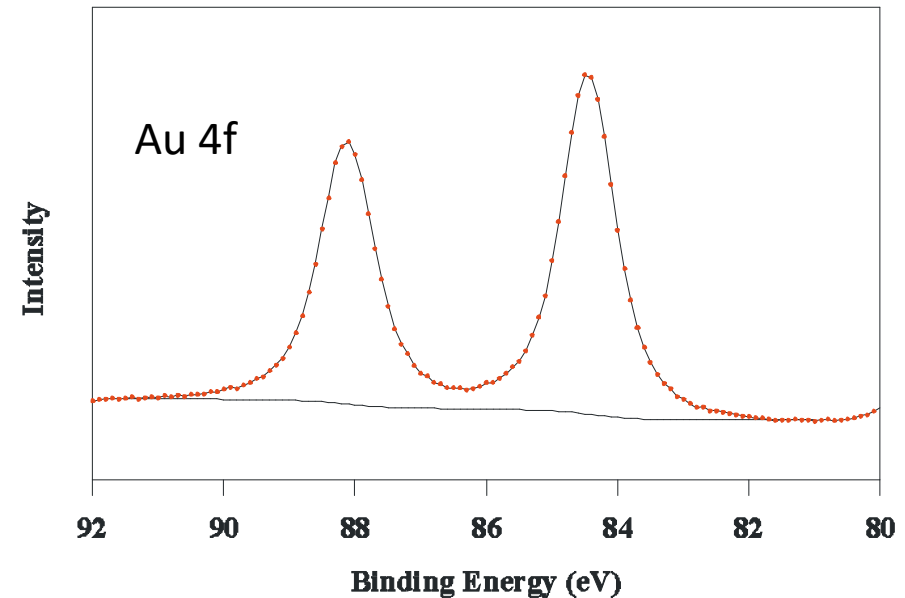
$$\text{Atomic \%} = \frac{\frac{I_A}{S_A}}{\sum_i \frac{I_i}{S_i}} \times 100\%$$

Peak Area of element A

Sensitivity factor of element A

Peak Areas / Sensitivity factors of all other elements

Peak Area measurement
Need background subtraction



Thank you